



GREAT ZIMBABWE UNIVERSITY

MUNHUMUTAPA SCHOOL OF COMMERCE

DEPARTMENT OF BANKING AND FINANCE

**THE ADOPTION OF THE CRYPTOCURRENCIES AND ITS EFFECT IN THE
MODERN ECONOMY: A CASE OF THE ZIMBABWEAN ECONOMY**

DISSERTATION

BY

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**IN PARTIAL FULFILMENT OF A MASTER OF COMMERCE DEGREE IN
FINANCE**

MASVINGO, ZIMBABWE

YEAR 2020

DECLARATION

I, Gerald Hlabiso, I declare that am a bonafide producer and owner of this research paper and the work presented therein is my own. I do also affirm that it has not been submitted by any other student to any academic institution. The contents of this paper have been submitted in partial fulfilment of the **A MASTER OF COMMERCE DEGREE IN FINANCE**

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I dedicate this research thesis to the Hlabiso Family. We have Mastered it!

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ABSTRACT

The purpose of the study was to investigate the effect of the adoption of cryptocurrency to the modern economy. A case of the Zimbabwean Economy. Virtual currencies have become the new normal in the financial space particularly the crypto currencies. Although much has been written about the significance of crypto-currencies the world over much research has not been done to establish the impact of cryptocurrency in Zimbabwe because of its ban in 2018. The study sought to found out the benefits of cryptocurrencies adoption in the economy, To investigate the regulatory framework that needs to be put in place for crypto currency operation in Zimbabwe, to find out if cryptocurrency can solve the liquidity crunch and the higher cost of international remittances and to analyse the opportunities and threats that cryptocurrencies poses to the economy. The study employed a positivistic research approach which made use of quantitative deductive approach. The study made use of 323 respondents where data was collected using interviews and questionnaires. The study concluded that cryptocurrency adoption have benefits that it brings to the economy. Therefore government have to legalise the use of the currency. Due to the characteristic of the currency it have been found out that cryptocurrency can solve the liquidity crunch in the economy and lowers the cost of remittances. The study has also found out that cryptocurrency brings opportunities such as lessen the effects of inflation, easier of transacting, fast transactions, financial inclusion and divisibility of the currency. The study recommends that the government have to issue laws that governs cryptocurrency operation in the country. The Reserve Bank have to come up with a policy document that clearly mentions how the cryptocurrency are to be recognised (digital currency or virtual currency), how it has to be used in the economy, the guidance of how the initial coin offerings have to be done and the laws that covers international remittances. The government have to include the use of cryptocurrency as taxable transaction. The cost of remittances have been reduced by around 7% if the money is remitted by blockchain technology rather than the normal money service remittance. This has therefore reduced the cost of remittance significantly. The government has to allow cryptocurrency remittances since a significant population of Zimbabweans reside in other countries. The government has to allow the trading of cryptocurrency in the economy since it bring more opportunities than threats in the economy.

TABLE OF CONTENTS

DECLARATION	i
RELEASE FORM	ii
APPROVAL FORM.....	iii
DEDICATION	iv
ACKNOWLEDGEMENTS	v
ABSTRACT.....	vi
TABLE OF CONTENTS	vii
LIST OF TABLES	xi
LIST OF FIGURES.....	xii
CHAPTER I.....	1
INTRODUCTION.....	1
1.0 Introduction	1
1.1 Background to the Study	1
1.3 Statement of the Problem	5
1.4 Research Objective	6
1.4.1 Main Research Objective	6
1.4.2 Sub- Research Objectives	6
1.5 Research Questions.....	6
1.5.1 Main Research Question.....	6
1.5.2 Sub Research Questions.....	6
1.6 Conceptual Framework	6
1.8 Significance of the Study.....	9
1.9 Delimitation of the Study	10
1.9.1 Geographical Delimitation	10
1.9.2 Theoretical Delimitation	10
1.10 Limitation of the study	10
1.11 Assumption of the Study	11
1.12 Definition of Key Terms	11
1.12 Structure of the study	11
1.13 Summary	12
CHAPTER II.....	12

LITERATURE REVIEW	12
2.0 Introduction	12
2.1 Conceptual Definitions.....	13
2.1.1 Cryptocurrency	13
2.1.2 Alt-coins.....	15
2.2 The benefits of cryptocurrency in an economy	20
2.2.1 Regulatory Framework for Cryptocurrency	22
2.2.2 Cryptocurrency, liquidity and cost of international remittances	25
2.2.3 Opportunities and threats of cryptocurrencies in the economy.....	28
2.3 Theoretical Framework	31
2.3.1 The Diffusion of Innovation Theory.....	31
2.3.2 Disruptive Innovation Theory	34
2.4 Empirical Literature Review	36
2.5 Research Gap	37
2.6 Summary	38
CHAPTER III	38
RESEARCH METHODOLOGY	39
3.0 Introduction	39
3.1 Research Philosophy	39
3.1.1 Research Approach.....	40
3.1.2 Research Strategy	40
3.1.3 Time Horizon	40
3.2 Research Design	40
3.3 Research Design	41
3.4 Target Populations	41
3.5 Sample Size	42
3.5 Sampling Techniques.....	43
3.6 Research Instruments	43
3.7 Data Sources.....	44
3.8 Data Collection Procedures	45
3.9 Validity and Reliability	45
3.9.1 Validity	46
3.9.2 Reliability.....	46

3.10 Ethical Consideration	47
3.11 Data Presentation and Analysis	47
3.11.1 Multiple Regression Assumptions.....	48
3.12 Pilot Study.....	48
3.13 Summary	48
CHAPTER IV	49
DATA PRESENTATION, DISCUSSION AND ANALYSIS.....	49
4.0 Introduction	50
4.1 Response Rate.....	50
4.2 Study Demographics	51
4.2.1 Gender of the respondents.....	51
4.2.2 Level of Education.....	51
4.2.3 Number of Years Working in the Organisation	52
4.3 Blockchain Knowledge	53
4.3.1 Type of Blockchain Technology	54
4.3.2 Motive for using the Blockchain Technology.....	55
4.3.3 Satisfaction of the Technology.....	56
4.4 Descriptive Statistics	57
4.4.1 Benefits of Cryptocurrency in the Economy.....	57
4.4.2 Creating a Regulatory Framework.....	60
4.4.3 Cryptocurrency, liquidity and cost of international remittances	63
4.4.4 Opportunities and threats of cryptocurrencies in the economy.....	67
4.5 Inferential statistics	70
4.5.2 Multiple Regression Analysis	71
4.5.3 Validity and Reliability of the Model.....	71
4.4 Results per Hypothesis.....	73
4.4.1 Cryptocurrency adoption.....	74
4.6 Summary	80
CHAPTER V	81
SUMMARY, CONCLUSION AND RECOMMENDATIONS	81
5.0 Introduction	81
5.1 Summary	81
5.2 Conclusions	81

5.3 Recommendations	84
5.4 Area of further study	85
REFERENCE LIST.....	85
APPENDIX.....	90
APPENDIX 1: COVER LETTER.....	90
APENDIX 2-Questionnaires for RBZ and Ministry of Finance Employees	91
APENDIX 3- Interview Guide	97

LIST OF TABLES

Table Number	Description	Page Number
Table 2.1	Top five cryptocurrencies by market capitalisation	16
Table 3.2	Target Population of the Study	42
Table 3.2	Sample Size	42
Table 4.1	Questionnaire and Interview Response rate	50
Table 4.2	Working Experience of Respondents	53
Table 4.3	Benefits of cryptocurrency in the Economy	57
Table 4.5	Creating a regulatory framework	61
Table 4.5	Cryptocurrency and Liquidity Crunch	63
Table 4.6	Cost of Remittance	66
Table 4.7	Opportunities and Threats of cryptocurrency adoption in an economy	68
Table 4.8	Skewness and Kurtosis statistics	71
Table 4.9	Validity data for the sample	72
Table 4.10	Reliability statics for model constructs	73
Table 4.11	Descriptive for study	73
Table 4.15	Correlation matrix for cryptocurrency adoption as a dependent variable	76
Table 4.16	Summary of statistics of regression model with adoption of cryptocurrency (AoC) as the dependent variable	76
Table 4.17	ANOVA results for regression model adoption of cryptocurrency (AoC) as the dependent variable	77
Table 4.18	Regression coefficient results and hypothesis decisions with adoption of cryptocurrency (AoC) as a dependent variable	78

LIST OF FIGURES

Figure Name	Description	Page Number
Fig 1.1	Total Market Capitalisation of Cryptocurrencies from 2014-2020	2
Fig 1.2	Conceptual Framework	8
Fig 2.2	Early use of Bitcoin clustered around gambling and the dark web	17
Fig 2.3	The Diffusion Innovation Theory	33
Fig 2.4	Model of Disruptive Innovation	35
Fig 4.1	Gender of the respondents	51
Fig 4.2	Gender of the Respondents	52
Fig 4.3	Blockchain Technology Knowledge	54
Fig 4.4	Most popular/ used Blockchain Technology	55
Fig 4.5	Motives for using the Blockchain Technology	56
Fig 4.6	Satisfaction Level after Technology Use	
Fig 4.8.	Independent predictor variables of cryptocurrency adoption in the hypothesised model	75

CHAPTER I

INTRODUCTION

1.0 Introduction

This study focuses on the adoption of cryptocurrencies and its effects on the modern economy. Cryptocurrencies and the concept of digital currencies have become one of the most anticipated topics of recent years. According to Jonh (2017) from the year 2010 to year 2017, it is said that cryptocurrencies had a total market capitalisation of approximately \$205 billion and a daily trade volume of more than \$10 billion. Though cryptocurrencies are dominating the market and the new normal, the Zimbabwean government had put a crypto ban, though peer to peer trading continues to thrive in the economy. This study advocates for the government and the Reserve Bank of Zimbabwe to re-consider adopting cryptocurrencies and putting in place robust regulatory framework that allows cryptocurrency to operate in the economy. The study explores the benefit of the cryptocurrencies in the economy and to propose a regulatory framework for the digital currencies to prevail in the economy.

1.1 Background to the Study

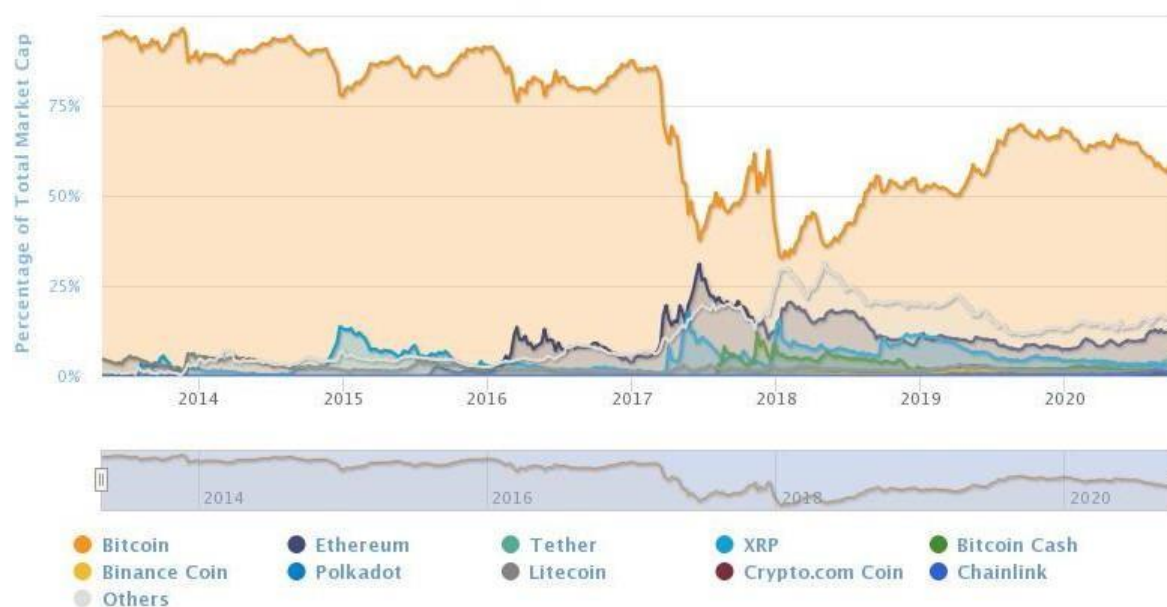
Despite the government's 2018 crypto ban, peer to peer trading continues to thrive in Zimbabwe, as the country faltering monetary policies seemingly push investors towards digital currencies. The adoption of cryptocurrency is seen to be on the rise in Africa, witnessed by a greater crypto ownership, trade and preparations for regulation frameworks. A report published by Arcane Research and Luno (2020), found that African nations such as Uganda, Nigeria, South Africa, Ghana, Kenya and Zimbabwe are frequently among the top 10 countries to Google Search about Bitcoin. A report titled A of Crypto in Africa (2020) recognises Africa as one of, if not the most promising continents for adoption of cryptocurrencies. The report observed that African countries share key similarities and trends despite the continent being so diverse, economic problems ranging from high inflation rates and currencies to financial issues such as capital controls and lack of banking infrastructure create a fertile ground for alternate developments.

Cryptocurrency originated with the white paper that was published in 2008 under the pseudonym 'Satoshi Nakamoto'. It is a digital or virtual currency that is secured by cryptography which makes it nearly impossible to counterfeit or double spend. Many

cryptocurrencies are decentralized networks based on block chain technology- distributed ledger enforced by a disparate network of computers and a defining feature of cryptocurrencies is that they are generally not issued by any central authority, rendering them theoretically immune to government interference or manipulation (Frankenfield and Sonnenshein, 2020). The first cryptocurrency was Bitcoin, which still remains the most popular and most valuable. Today, there are thousands of alternate cryptocurrencies with various functions and specification. Some of these are clones or forks of Bitcoin, while others are new currencies that were built from scratch.

As at September 2020, there were over 18.5 million Bitcoins in circulation with a total market capitalisation value of around \$195.6 billion and 18.5 million in circulation (Coinbase, 2019). Some of the competing cryptocurrencies spawned by Bitcoin success, known as Altcoins include Litecoin, Peercoin, and Namecoin, as well as Ethereum, Cardano and EOS. Today, the aggregate value of all cryptocurrencies in existence is around \$214 billion- Bitcoin currently representing 68% of the total market capitalisation value (CoinMarketCap, 2020). The Fig 1.1 below shows the total market capitalisation for the past 6 years for cryptocurrencies

Fig 1.1 Total Market Capitalisation of Cryptocurrencies from 2014-2020



Source: Coinmarketcap.com (2020)

Cryptocurrencies has become the new normal, first and foremost thing which was included in its features of Bitcoin is decentralisation, which means there is no central authority or any financial institution that can control its transaction network. It implies that it is under control

of any agency which can play with monetary policy rather it is controlled by a protocol maintained by peer to peer system (Sharma and Sharma, 2018). It is easy to set-up since it is an open source software-based payment system, which payments are recorded in a public ledger using its own unit of account called the Bitcoin. Bitcoins are stored in wallets with digital credential for an individual Bitcoin holding which enhance anonymity and transparency since the wallets uses public-key cryptography in which two keys are generated one public key and one private key which are generated. The beautiful thing about Bitcoin either national or international transactions, it does not involve any charges to its users, thus it avoids double spending by verifying each transaction added to its chain for uniqueness in the inputs (Bornholdt and Sneppen, 2014).

Globally, some countries have taken steps to regulate cryptocurrency at a legislation level. The Virtual Currency Act, 2017, promulgated in April 2017 by the Japanese government perhaps the best evidence this; while Australia's crypto currency exchange providers are now subject to regulation under the Anti-Money Laundering and Counter-Terrorism Financing Act, 2006. On a federal, the United States Internal Revenue Services (IRS) officially classified cryptocurrency as a property for tax purpose in 2014 (Internal Revenue Services, 2017) and passed the Uniform Regulation of Virtual Currency Business Act, 2017. On state level, the New York Department of Financial Services has implemented a regulatory framework for crypto-currencies exchanges. In 2014, the Canadian government passed Bill C-31 amendments made to the Proceeds of Crime (Money Laundering) and Terrorist Financing Act (PCMLTF), which seeks to address the regulation of crypto assets exchanges and payment processors. All these regulations are put in place for the proper functioning of the digital currencies; thus, Zimbabwe has to mirror these regulations to pave way for the smooth running of the virtual currencies.

In Africa the journey has come a long way in establishing itself in the block chain and crypto currency space. Despite a slow start, the continent is slowly catching up with the rest of the world. By the end of 2019, there was a general call for regulation on the crypto-currency market across the globe. In Africa, however, different countries have different reactions to this call. Some countries have managed to formulate a regulatory framework, some have declared a total ban while others are taking a rather slow approach. Examples of African countries with a positive outlook on cryptocurrency are South Africa, where the South Africa Reserve Bank published a document titled "Consultation on Policy Proposals for Crypto Assets" which authorities sought to have a comprehensive set of regulations created for the cryptocurrency

space. The South Africa Revenue Services (SARS) had announced they would be taxing income derived from crypto-related activities and further drafting a crypto lawdraft. In Kenya, the president assented the Finance Bill of 2019, under the new bill, gave the taxman power to tax digital currencies. Nigeria has not be sparred out, In 2018 Nigerian Securities and Exchange Commission (SEC), set up a Fintech Road Map Committee tasked with the duty to develop a framework for the regulation of Virtual Financial Assets (VFAs) and their exchanges in Nigeria.

However some countries in Africa has put a ban on the cryptocurrency trading. In Burundi 2019, the country officially banned all forms of transaction involving Bitcoin and other digital currencies. The purchase, sale, use, and holding of so-called virtual currencies is prohibited in Algeria since 2018. The exchange office in Morocco issued a statement in which it declared that, the general public that are transacting via virtual currencies constitute an infringement of the exchange regulations which can be liable to penalties and fines. In 2017, the Bank of Namibia in a position paper announced that it considers the spending of cryptocurrency within the nation to be illegal and advised the Namibians to be wary of malfeasance and other risk factors, should they choose to trade in virtual currencies. In 2018, Egypts Dar al-Ifta, issued a religious decree classifying commercial transactions in Bitcoin as haram (prohibited under Islamic law).

Back home, after a study held by the Reserve Bank of Zimbabwe Fintech group in 2018, which looked into cryptocurrencies, the central bank categorically stated that cryptocurrencies are not welcomed in Zimbabwe. The RBZ made it clear that crypto were prone to endanger the public (RBZ, 2018). This came in an ongoing legal battle with Golix (Zimbabwe largest cryptocurrency exchange), however they seem to be a pro-crypto stance by the regulator. The Reserve Bank of Zimbabwe is considering a regulatory sandbox for cryptocurrency stance. The proposed sandbox is reportedly part of the border agenda by the authorities to revamp the nation's financial sector, which has seen massive hyperinflation since 2007 (Mutepfa, 2019). Meanwhile, Zimbabwe government is also partnering with Apollo Fintech to develop a gold backed cryptocurrency for the country, a collaboration which may allow Zimbabweans to have access to the Apollo wallet for digital currency payments

Despite the government 2018 crypto ban, peer to peer trading of cryptocurrencies continues to thrive in Zimbabwe, as the country's faltering monetary policies seemingly push investors towards digital currencies. In mid-2019, Cryptomania reports that in Zimbabwe the price of

the Bitcoin reached a 600% premium and in 2017 the rate of Bitcoin (XBT) to Zimbabwean dollar reached its peak. Cryptocurrencies are closer to in from to physical currencies than other virtual currencies because they are also a medium of exchange for physical assets. Harvey (2015) cited in Mazikana (2019) argues that most of the modern world's money supply is in digital form thus crypto presents a greater socio-economic inclusiveness through enhanced financial security.

A report realised by Ecobank on the state of cyprocurrency regulation in Sub-Saharan Africa in 2018 showed that of the 38 countries studied, only two, South Africa and Swaziland have favourable or official permissive stance on cyptocurrecies. While Zimbabwe, Burundi, Morroco, Algeria and Namibia have taken in outlawing digital currencies. As the report by EcoBank (2018) puts it; "African governments and central banks are mostly adopting a 'wait and see' approach when it comes to the regulating crypto currencies". Rather than being the first mover, it adds "African countries appear to be looking to their neighbours to regulate and innovate first, and learn from their mistakes." Like most of the African governments which are seeming to be bidding on their time, in Zimbabwe crypto currency trade has notably risen due to the countys economic crisis as local continue to turn to bitcoin as a store of value. Indeed, for a brief period, Zimbabwe had the highest bitcoin prices in the world. Therefore this study advocates for the government of Zimbabwe to adopt and regulate the cryptocurrency industry.

1.3 Statement of the Problem

Virtual currencies have become the new normal in the financial space particularly the crypto currencies. Although much has been written about the significance of crypto-currencies the world over much research has not been done to establish the impact of cryptocurrency in Zimbabwe because of its ban in 2018. The literature on crypto currencies has proved that there is a research gap on the impact of the adoption of crypto currencies and its effect on the economy. The effects of crypto currencies and the feasibility of it through the use by banks and individual is currently unknown. It is the aim of the study to close the knowledge gap, by showing how the emergency of crypto often referred to as disruptive technology create a bridge between information technology and finance because changing the way some institution works. The study will explore the feasibility of the crypto currency in the Zimbabwe economy, regulatory framework that needs to be put in place for its smooth operation, the effect of

cryptocurrency to cure economic problems such as financial issues like liquidity problems and the opportunities and threats that come after the adoption of cryptocurrency in the economy.

1.4 Research Objective

1.4.1 Main Research Objective

- To analyse impact of the adoption of cryptocurrencies in Zimbabwe and its effect on the economy

1.4.2 Sub- Research Objectives

- To explore the benefits of cryptocurrencies in the economy
- To investigate the regulatory framework that needs to be put in place for cryptocurrency operation in Zimbabwe
- To find out if cryptocurrency can solve the liquidity crunch and the higher cost of international remittances
- To analyse the opportunities and threats that cryptocurrencies poses to the economy

1.5 Research Questions

1.5.1 Main Research Question

- What is the impact of the adoption of cryptocurrencies in Zimbabwe and its effect on the economy?

1.5.2 Sub Research Questions

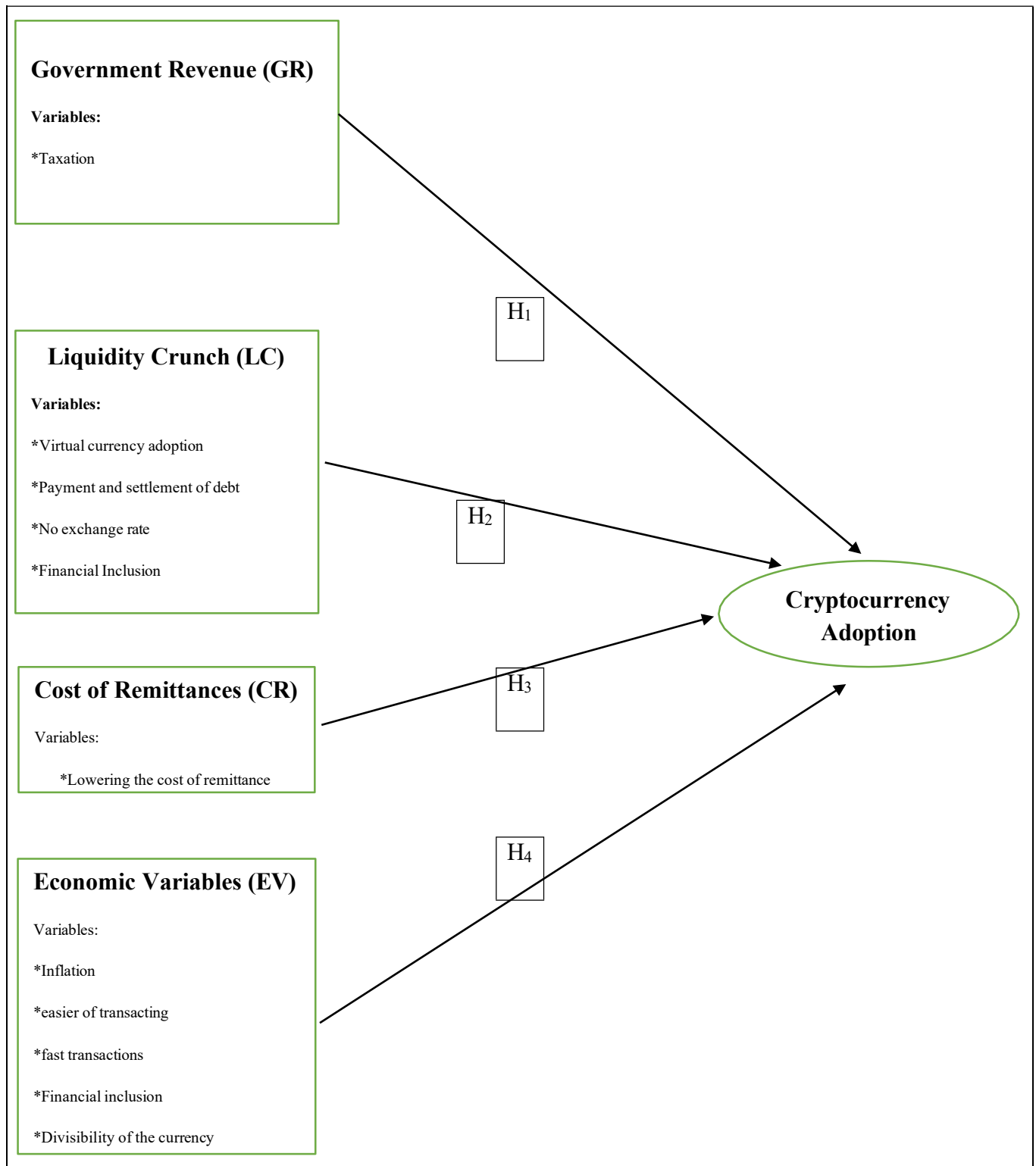
- What are the benefits of cryptocurrency in the economy?
- What is the regulatory framework needed to be put in place for cryptocurrency operation in Zimbabwe?
- Can cryptocurrencies solve the liquidity crunch and higher cost of remittances in Zimbabwe?
- What are the opportunities and threats that cryptocurrency pose to the economy?

1.6 Conceptual Framework

According to Adom *et al* (2018), a conceptual framework explains the path of a research and grounds it firmly in theoretical constructs. The overall aim of the theoretical framework is to make the research findings more meaningful, acceptable to the theoretical constructs in the research field and ensures generalizability. The research made use of four research constructs

which made the independent variables of the research that is government revenue (GR) which has the taxation variable in it; liquidity crunch (LC), which had virtual currency adoption, payment and settlement of debt using virtual currencies, no exchange rates and financial inclusion; cost of remittance (CR) which had lowering of cost of remittances as a variable and economic variables (EV) which included variables such as inflation, easier usage and swift transactions, financial inclusion and divisibility of the currency. Fig 1.1 shows the conceptual framework of the study.

Fig 1.1 Conceptual Framework



1.7 Statement of Hypothesis

The research hypothesis is a testable proposition or predictive statement about the possible outcome of the study. The research hypothesis are to assist the research questions and the conceptual framework to form the relationships that the study sought to find. The study hypothesis are listed below

1.7.1 Main Research Hypothesis

H₁- There is a relationship between the adoption of cryptocurrency and economic growth

1.7.1.1 Specific hypothesis

H₁- There is a positive relationship between cryptocurrency adoption and an increase in government revenue

H₂- There is a negative relationship between cryptocurrency adoption and liquidity crunch

H₃- There is a negative relationship between cryptocurrency adoption and cost of remittances

H₄- There is a positive relationship between cryptocurrency adoption and economic variables

1.8 Significance of the Study

The encouragement of the adoption of cryptocurrency in Zimbabwe is a crucial in view of many challenges which the country is experiencing from liquidity challenges to inflation. This study advocates for the adoption of crypto currency, to gain an understanding of the crypto-currencies operations and the need of the crafting of the regulatory framework for the smooth running of crypto-currency operations. This study would be of relevance to policy makers in knowing the feasibility of the cryptocurrency in the country, how it can solve the liquidity crunch, as a consideration for as an investment destination and a promulgation of a regulatory framework for cryptocurrency in Zimbabwe. Academics would benefit from the findings of the research as it brings forth new knowledge on cryptocurrency adoption in the country, an aspect that is currently shallowly published on. This thesis brings new insights into the adoption of cryptocurrency and its effect on the modern economy.

1.9 Delimitation of the Study

According to Theofanidis and Fountouki (2018), delimitation of the study are those characteristics that limit the scope and define the boundaries of the study. Therefore study will be delimited in the following aspects:

1.9.1 Geographical Delimitation

The study focused on all the crypto currency firms in major cities of the counter such as Harare, Bulawayo, Mutare, Gweru, Kwekwe and Masvingo. The researcher focused on major cities because in the major cities that is where cryptocurrency is most found. The urban people are mostly the ones who are knowlegable about cryptocurrency than rural folks. The researcher delimited to major cities because of the limited time frames to carry out the research and the research budget thus the researcher concentrated on major towns.

1.9.2 Theoretical Delimitation

The researcher used literature that only focused on cryptocurrency. Therefore the researcher focused mainly on literature from 2015 to 2021, which would give the researcher a proper research gap for the study.

1.10 Limitation of the study

The study will be limited to the finance industry in Zimbabwe. That is major cities would be included in the study thus it would not be reflective of the global reach and business environment. The researcher in vein was mirroring the practices of some of the cryptocurrency activities globally through the use of secondary data. The researcher had limited time frames to carry out the research, hence the researcher made use of the drop and pick approach in the distribution of questionnaires and scheduled interviews which some respondents were unavailable, feared for their schedule and hiving tight work schedules. The researcher made use of the drop and pick approach to avoid pressuring the respondents and a letter from the university which clearly stated that the research was strictly for educational purposes and respondent's privacy is protected was used. The respondent was assured of their withdrawal from the researcher if the concern arise and not to include their names in filling their questionnaires.

Due to the global pandemic of the novel COVID-19 virus, the government introduced a locked down and strict social distancing measures. This made the movement of the researcher limited and sometimes the respondents were not available at their work places. This therefore resulted

in a sample size being smaller than expected. The researcher made use of an online data collection tool called survey planet to distribute the questionnaires and interviews were scheduled through Skype and Zoom which made the respondents reachable.

1.11 Assumption of the Study

In conducting the study, it was assumed that

- The respondents of the study provided unbiased data that is reliable for the study
- The sample size used in the research study is representative of the entire population
- The questionnaires were administered and answered within a short period of time
- The data collection methods and instruments were reliable and valid thus gives a reliable and accurate result for the study.
- All ethical considerations were considered in the study

1.12 Definition of Key Terms

Cryptocurrency: A cryptocurrency (or crypto currency or crypto for short) is a digital asset designed to work as a medium of exchange wherein individual coin ownership records are stored in a ledger existing in a form of computerized database using strong cryptography to secure transaction records, to control the creation of additional coins, and to verify the transfer of coin ownership. It typically does not exist in physical form (like paper money) and is typically not issued by a central authority.

Peer to Peer Trading: Is an electricity trading with a business model based on interconnected platforms that serves as an online marketplace where consumers and producers meet to trade electricity directly without the need for an intermediary.

Virtual Currency: Is a type of unregulated digital currency, which is issued and usually controlled by its developers and used and accepted among members of a specific virtual community.

Digital currency: Is a balance or a record stored on the Internet, in an electronic computer database, within digital files or within a stored value card.

1.12 Structure of the study

The research was carried out in five different chapters. In chapter one contained an introduction to the research were the research background, statement of the problem, research objectives, research questions, significance of the study, delimitation of the study and limitation of the

study was explained. Chapter 2 presented the literature review of the study. The literature was silted into 3 section that is the conceptual literature revive, theoretical literature review and empirical literature review. The research indicated the research gap that the research intends to explore. Chapter 3 presented the research methodology used in the study. This chapter explored the research approach used, target population, sampling techniques and methods used and indicating the ethical consideration taken in carrying out primary data. Chapter 4 presented data analysis where data collected was analysed using the SPSS software. The study made use of multi linear regression in data analysis. Bar graphs, pie charts and tables where used for data presentation. Chapter 5 discussed the research findings, gave conclusions and recommendations of the study. Areas of further study were outlined at the end chapter.

1.13 Summary

The chapter provided an introduction to the study, the background of the study, and the problem statement. This then was followed by the research objectives and questions which the study seeks to explore, the later part of the chapter provided the significance of the study, delimitations, limitations in which contingency plans were provided. The next chapter will provide a critical literature review related to the study with the main focus on the research questions, giving a research gap that the study seeks to cover.

CHAPTER II

LITERATURE REVIEW

2.0 Introduction

2.1 Conceptual Definitions

This section defines the conceptual terms of the study, that is the independent and dependent variables of the study. This section will define cryptocurrency and how it is operating in the modern economy globally. This section would define, make an analysis and conduct the applicability of the concepts to the present study.

2.1.1 Cryptocurrency

Bitcoin originated with the white paper that was published in 2008 under the pseudonym “Satoshi Nakamoto”. It was published via a mailing list cryptocurrency and has a similar appearance to an academic paper (Nakamoto, 2008). The creators’ original motivation behind Bitcoin was to develop a cash-like payment system that permitted electronic transaction but also included many of the advantageous characteristics of physical cash. The emergency of the crypto currencies and block chain technologies is part of a broader wave of technologies that facilitate peer-to-peer (P2P) commerce, individualization of products, and flexibilization of production methods. For a variety of reasons, this wave gained traction after the global financial crisis a decade ago. Large digital platforms, such as Alibaba, Amazon, Uber, and Airbnb, are replacing many brick and mortar stores, services companies, and long term employment relationships.

Block chain technologies aim to go one step further. They organise P2P transaction and P2P information flows without companies that operate digital platforms. Whether these technologies will completely eliminate middleman or whether new forms of trusted intermediaries will emerge remains unseen.

Cryptocurrencies are the first and therefore the most developed application of blockchain technology that creates money without the central bank and facilitate payments without financial institutions. The success of several cryptocurrency put competitive pressure on transaction methods by existing financial institutions. However serious limitations have become apparent. Decentralized organisation of markets without trusted intermediaries can be very costly, and the volatility of the value of cryptocurrencies is a big obstacle to their becoming an alternative to legal tender. The emergency of blockchain technologies has triggered a flurry of activities in Europe and Central Asia, where people use cryptocurrency for cross border transactions and providing blockchain services. Government are also experimenting on the block chain technologies to make their services more secure and more transparent.

According to the World Bank (2020), 10 years after an ingenious experiment to create a cryptocurrency that allows secure and anonymous digital transactions to take place without the involvement of the central bank or commercial banks, cryptocurrency have become a multimillion dollar industry. By December 2017, the average price of one Bitcoin (the first cryptocurrency) had risen from just a few cents in 2009 to \$1500, doubling its value in a single month. These gains attracted many investors across the world. On December 1; 2017 the U.S Commodity Futures Trading Commission approved trading in Bitcoins futures. Although the price had declined to about \$8000 in April 2018, the value of the Bitcoin in circulation was about \$150 billion as of April 2018. Big companies, and individuals working together in large pools, are competing for the right to add new transactions to the existing chain of transactions. Their revenues in the form of new Bitcoins and transaction fees, are close to \$20 million a day. In the wake of Bitcoins success, hundreds of alternative cryptocurrencies have been created. Digital tokens have been issued as a general currency; for specific purposes (for example, to rent computer capacity or cloud storage); and as an alternative to traditional shares in companies.

Cryptocurrency have evoked strong reactions. Critics call these virtual currencies a bubble, a scam, and even evil (Krugman, 2013; Popper, 2018). Supporters predict that cryptocurrency will ultimately replace money (Rooney, 2018). There is less disagreement about the underlying blockchain technology, a common database, stored in multiple locations. Many recognise that the blockchain protocol can lead to tamper proof, secure information systems without the need for a single administration (Rooney, 2018). The believers of cryptocurrency foresee utopian societies of self-regulating individuals, without government or trusted intermediaries. Doubters argue that the number of useful applications has been exaggerated, that lack of regulation can have disastrous effects, and that in most cases trusted intermediaries will continue to provide useful services.

Jonhson (2018) argues that it is unclear how these technologies would develop in the long run, Conceivably, they could be absorbed by existing institutions, with central banks issuing digital cash, governments using blockchain to maintain information systems, and commercial banks putting payment systems on the blockchain. Many intermediaries might become absolute, and many new financial instruments might be created by companies that do not yet exist. The main legacy of cryptocurrency may not be the blockchain technology but standardized digital IDs using combination of public and private keys on open-source software (Clark, 2017). Such

development would allow individuals to own more of their data, instead of participating in proprietary information networks (Jonhson, 2018).

Cryptocurrency and blockchain activities are widespread in Europe and Central Asia (ECA). Massive mining of cryptocurrencies takes place in Iceland, Sweden, and Georgia. Many Russian own digital wallets, and experiments are ongoing in Serbia and Tajakistan to use blockchain technology to make the sending of remittances more efficient (UNDP, 2018). Estonia is using blockchain software in registries and plans to extend its use to medicine (<https://e-estonia.com/>). Start-ups in many countries in ECA are contributing to these technologies, attracting finance for their activities via initial coin offering (ICOs) (Ben, 2017). Household investments in cryptocurrencies are not insignificant. Switzerland aims to becoming a cryptocurrency and blockchain hub and is leading in adjusting regulations to these new technologies.

However the use of cryptocurrencies and blockchain technologies pose a range of policy challenges. They include the need to (a) apply rules of financial oversight, consumer protection, and tax administration while at the same time encouraging and facilitating innovation; (b) deal with the massive volume of electricity used to mine cryptocurrencies; and (c) to determine whether governments and central bank can use blockchain technologies to improve their services. Therefore the policy marker has to strike a balance that curbs the hype and unleashing potentially transformational new opportunities. There is a need for international coordination to share the best practices, avoid regulatory arbitrages and explore how to regulate global decentralized networks.

2.1.2 Alt-coins

Bitcoin has spurred a variety of copy-cats as well as novel cryptocurrencies that act more as platforms than mere currencies. The cryptocurrency industry has spanned to Altcoins such as Ethereum, Ripple, Bitcoin Cash and Litecoin which are extensively dominating the market like Bitcoin. What is common amongst these, and therefore what makes them cryptocurrencies is their distributed nature, use of cryptograph to ensure integrity, and the use of the underlying blockchain technology to order instructions. Variations include the consensus mechanism used, the cryptographic algorithms used, the math problem that must be solve, value and nature of awards, privacy mechanism, and block solution time amongst others. The open-source nature of the programmes that Bitcoin lends itself to competition along the lines of Hayek's theory of competitive currencies (Ametrono, 2016). According to Bjerg (2016), since the barriers to entry

are low, competing currencies can and have emerged and this implies that Bitcoin existence is by its nature dependent on its ability to competitively service its customers.

Table 2.1 shows the market capitalisation of Bitcoin and other Altcoins indicating the dominance of cryptocurrency in the modern world.

Table 2.1 Top five cryptocurrencies by market capitalisation

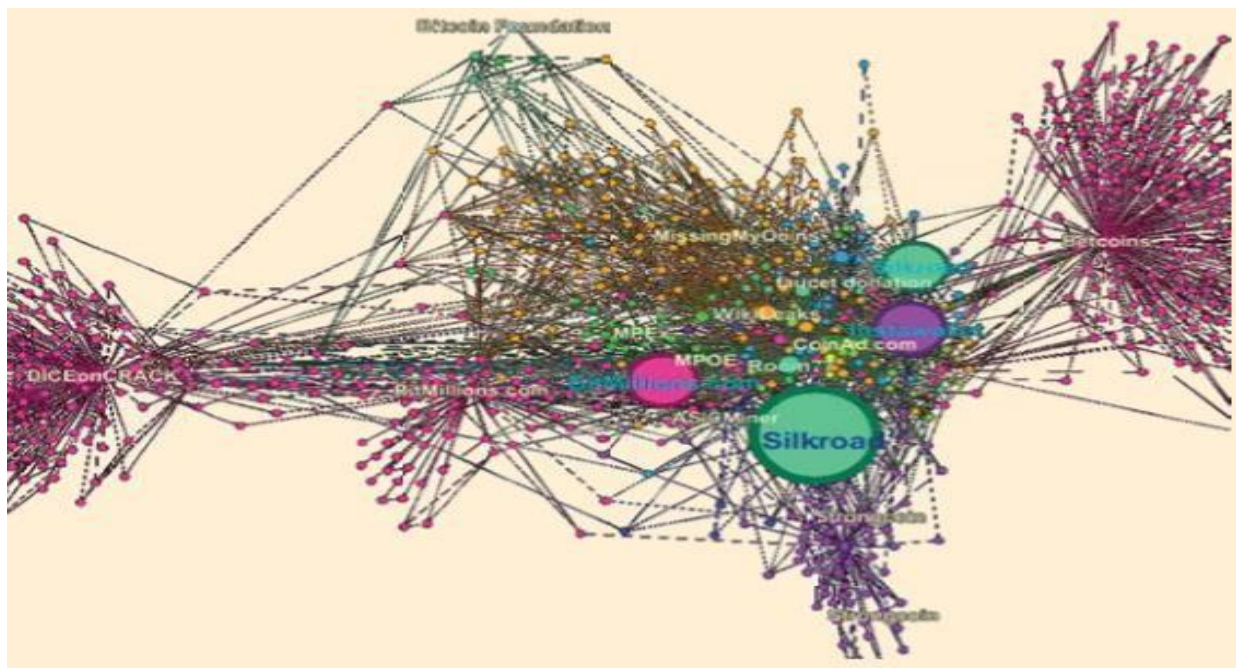
# ▲	Name	Price	24h	7d	Market Cap ⓘ	Volume ⓘ	Circulating Supply ⓘ
☆ 1	 Bitcoin BTC	\$31,504.27	▲ 0.67%	▼ 4.12%	\$586,377,940,278	\$78,072,761,225 2,478,164 BTC	18,612,650 BTC
☆ 2	 Ethereum ETH	\$1,296.70	▲ 2.18%	▲ 3.70%	\$148,389,066,461	\$39,092,691,514 30,147,862 ETH	114,436,049 ETH
☆ 3	 Tether USDT	\$1.00	▼ 0.07%	▲ 0.14%	\$25,428,456,700	\$108,069,710,011 107,956,787,089 USDT	25,401,886,298 USDT
☆ 4	 Polkadot DOT	\$17.25	▲ 9.51%	▲ 3.40%	\$15,607,192,213	\$2,614,060,390 151,580,297 DOT	905,006,952 DOT
☆ 5	 XRP XRP	\$0.2567	▲ 0.10%	▼ 10.11%	\$11,655,857,848	\$2,889,286,984 11,254,878,935 XRP	45,404,028,640 XRP

Source: Adapted for CoinMarketCap, 2021

According to Raynor de Best (2021), as of January 2021, the Bitcoin market capitalisation has reached an all-time high and had grown by over 400 billion U.S dollars when compared to the other previous months. The volume of Bitcoin in circulation currently sits on \$78 billion U.S dollars followed by Ethereum at \$39 billion, Tether at \$108 million, Polkadot at \$2 billion and XRP at \$2 billion. While the market capitalisation and cryptocurrency is circulation is growing at an unbelievable rate from 2013, the public is slowly becoming aware of its existence. As many as 25% of Americans were still not familiar with Bitcoin and as of 2019, while there are over 18 billion cryptocurrency in circulation, there are only 6.674 Bitcoin ATMs around the world thus the physical appearance of cryptocurrency is minimal, and the majority of ATMs are located in the US. Thus the calls of the study for African countries to accept cryptocurrency in their economies.

2.1.3 The usage of Cryptocurrencies to Modern economies

The use of cryptocurrencies has been dominate in many European and Central Asian countries which have provided a fertile ground for cryptocurrencies and blockchain technologies, especially since late 2016 (Amy, 2017). When cryptocurrencies emerged, almost 10 years ago, activities were small-scale. As everything else in the world, early transactions were used largely for gambling or for the purchase of illegal products on the dark web (Silk Road, 2014). Figure 2.2 shows the early use of Bitcoin clustered around gambling and the dark web.



Source: Source: blockchain.info.

The use of cryptocurrencies intensified at the end of 2016, especially for large cross-border transactions. When the prices of cryptocurrencies skyrocketed in 2017, investments in mining capacity increased sharply and people began investing in cryptocurrencies in the hope of benefiting from further price rises. Investments in blockchain technologies surged. Governments in many European and Asian countries began experimenting with the blockchain to improve their services. Some of the central banks are studying the issuance of legal tender in the form of digital currency and financial institutions are piloting blockchain applications to incorporate them in the existing financial architecture.

The European and Asian countries has become an important centre for the initial coin offering (ICOs). In terms of the number of projects, the Russian Federation ranks globally (with 8.8 percent of all projects), Switzerland fifth (6.9percent), Estonia seventh (3.0 percent), and Lithuania eighth (2.8 percent) (Figure 2.1). The reasons for strong interest in blockchain technologies in European and Asian countries are discussed below:

According to Aris (2019), blockchain technologies, which place a heavy emphasis on making financial intermediaries redundant, are particularly attractive in countries where trust in financial institutions is lacking, people want to avoid oversight, and or financial sectors are underdeveloped. Countries where corruption and political instability are higher, confidence in the rule of law is lower, and regulatory quality is lower tend to adopt Bitcoin more rapidly. A prime example is Republica Bolivariana De Venezuela, where people seek alternatives for bolivar, the value of which been eroded by hyperinflation. Nevertheless, some anecdotal evidence suggest that weak institutional or vulnerable banks are one reason behind the interest in cryptocurrencies (Dyussebekova, 2017). Financial dollarization remains substantial in the eastern part of the region, reflecting lack of trust in existing legal tender. It has resulted in much lower savings at banks than in other parts of the world (Gould and Melecky, 2017), thus households are looking for alternative saving options.

Another reason for the use of blockchain technologies is the desire to develop alternative means of transferring large funds. In Russia the largest issuer (more than \$956 million), followed by the United States (\$811million) and Switzerland (\$514 million), these funds were made available because of the TON blockchain (Khrennikov and Voltova, 2018). One of the goals of that ambitions project is to provide an alternative to the Swift international interbank payment system (Aris, 2017). Russia also has the largest number of users of the digital wallet on blockchain.com (UNDP, 2018). Despite these examples, it is doubtful that ICOs would have a broader application as venture capital if security is not built in for investors.

Reputable financial centers are striving to adjust to meet competition from disruptive technology like blockchain. Switzerland is leading in adjusting financial regulations to cover ICOs, ensuring that they are incorporated into the existing financial architecture rather than developed as an outside alternative (Atkins, 2018a; Financial Times, 2018). It aims to become a cryptocurrency and blockchain hub is reflected in its vibrant ICO activities. For example, Sirin Labs raised \$157 million for the development of a blockchain based smartphone. In line with these developments, a Swiss foundation, advised by Jacob Frenkel, chairman of JP

Morgans Chase International and Nobel laureate Myron Scholes, raised \$50 million to develop a cryptocurrency backed by Special Drawing Rights (SDRs). Saga would have a stable value and can be integrated into the existing financial sector, including anti-monetary-laundering checks, with deposits in the international Monetary Fund's SDR holding. France is also planning a regulatory framework for ICOs (Aris, 2017).

Most of the government in the European countries are accumulating in-house experience with blockchain pilots to improve government's services. Estonia, Georgia, and Ukraine have experimented with blockchain to set-up land and real estate registries. They are still searching for more specialized and more efficient designs, but the experiments have given a boost to efforts to digitalize government services. Some of the government banks are seeking improve their services through the use of blockchain technologies. The Russian state-owned VEB bank is piloting a new blockchain-based payment system with the regional government of Kiliningrad (Milano, 2018). Another state-owned bank, Sberbank, is partnering with Russia federal anti-monopoly service to use blockchain technologies to store and transfer documents.

Official bodies in the European and Central Asian countries are investing in blockchain research to improve services. The European Commission has funded a blockchain observatory to courage blockchain technologies and help formulate policy recommendations, especially for smart contracts and the improvement of government services (Young, 2017; Nicholson, 2018). Lithuania has opened a blockchain centre to incubate start-ups, partnering with similar centres in Melbourne and Shanghai (MediTelegraph, 2017). Separately, the central bank of Lithuania offers a one-year sandbox environment for start-ups that develop new digital financial technologies. Estonia is exploring opportunities to use blockchain technologies in medicine (<https://e-estonia.com>), whilst Georgia is investigating the possibility of supporting smart contracts. Serbia and Tajistan are experimenting with remittances on blockchain, in cooperations with the United Development Programme (UNDP, 2018). Azerbaijan is experimenting with digital IDs for banking using blockchain (Sputnik International, 2018). The Swedish central bank is considering launching its own digital currency (Aris, 2017).

Small European and Asian countries with a supportive business climate and the absence of legacy financial instruments are well placed to introduce new financial instruments based on blockchain technologies. Tokenization and ICOs enable small start-ups, which lack easy access to finance, to raise funds in global markets. Dynamic start-ups in the Baltic countries and several other small countries including Georgia, have issued ICOs. These examples are

instructive for other economies that have long been dominated by state-owned enterprises and have grown primarily through the non-tradable sectors. For many of those economies, the challenges is to unleash new growth potential in internationally competitive sectors. This new P2P technologies' provide a gateway to these markets. More specifically, activities on and contributions to blockchain networks are automatically exposed to international competition.

Seemingly more than in other parts of the world, governments are restraining natural monopolies of tech giants. People in the European and Asian region show strong privacy concerns when data become proprietary and are captured by tech companies. The open character of the blockchain architecture could break the monopoly on data. Several governments and the European Commission are looking at the possibility of using the new technologies to reduce the power of large digital network companies.

2.2 The benefits of cryptocurrency in an economy

Cryptocurrency is being adopted worldwide basing on the market capitalization of over \$78 billion equivalent to the United States Dollars (CoinMarketCap, 2021) thus economies around the world are starting to recognise its importance in the economy though it brings in some negative externalities in it acts. The optimistic view of cryptocurrencies use is backed by the fact that they make it easier to transfer funds between two parties in a transaction; these transactions are facilitated through the use of public and private keys for security purposes. These fund transfers are done with minimal processing fees, allowing users to avoid the steep fees charged by most banks. In addition, many economies have started to accept Bitcoin as a valid currency, especially economies that aim to get rid of cash have a very friendly approach to cryptocurrencies (Banjaku et al, 2018).

Like money uses in the economy, cryptocurrency have come with its benefits that are proving to be superior. According to Barros (2019), cryptocurrency provides a unit of measure that allows like traditional money the comparison of the value of different goods and services though they have a volatility that might prevent comparison at different times. Like traditional physical money it can be used as a means of payment and since they exist as global currencies, their value is the same anywhere in the world and no exchange rate is required in facilitation of international trade. Cryptocurrencies are divisible, like traditional currencies, into sub-units, which allows up to 8 decimals places. There are portability is even greater than physical money, being necessary only to have keys for access (Barros, 2019).

According to Banjakula et al (2018), cryptocurrency has provide a decentralization of transactions. Blockchain technology has revolutionized the way transaction are verified and managed. There is no central authority in the network, the network is distributed to all participants, and each computer mining Bitcoin is a member of this system. This means that the central authority in the network has no power to dictate the rules for owners of Bitcoin and even if some part of the network goes offline, the payment will continue to operate stable. Bitcoin also offers a peer to peer cryptocurrency network where there is no master server, which is responsible for all operations, thus the exchange of information (money) is between 2-3 or more software's clients. All installed by user program wallets are part of the Bitcoin network. Each client stores a record of all committed transaction's and the number of Bitcoin in each wallet. Transaction are made by hundreds of distributed servers. Neither banks or taxes, nor government can control the exchange of money between (Banjakula et al, 2018).

Cryptocurrencies uses the open code for mining cryptocurrency, the same algorithms that are used in online banking. However the only difference of internet banking is the disclosure of information about the users. All information about the transaction in the blockchain network shared (like how, when), but there is no data about the receipt nor the sender of the coins (there is no access to the personal information of the owners wallet. Thus blockchain transaction are fully anonymous and at the same time fully transparent (Li and Wang, 2017). According to Li and Wang, 2017 (2019), cryptocurrency offer transparency to its users. The Bitcoin stores the history of transactions that have ever taken place. It uses the sequential chain of blocks or blockchain, this keeps information about everything, thus if a company gas publicly used Bitcoin address, then no one will ever know that it belongs to the company. For complete anonymity companies usually use unique Bitcoin address for every single transaction.

According to Kampl (2014), cryptocurrencies usage in the economy stops inflation, that is, the maximum of coins is strictly limited to a certain number, there are neither political forces nor corporations that are able to change that order, thus there is no possibility for the development of inflation in the system. Cryptocurrencies have unlimited amounts of transaction's that is, each of the wallet holders can pay to anyone, anywhere and any amount. The transaction can be controlled or prevented thus a user can make transfers anywhere in the world wherever another Bitcoin wallet is located. That it offer no transaction boundaries. The payments made in the system are impossible to cancel, the coins cannot be faked, copied or spent twice thus these capabilities guarantee the integrity of the entire system and allowing online shops, resources and companies to accept Bitcoin globally (Jonker, 2017).

Fahmy (2018) argue that cryptocurrencies are easier to register and to use taking to account the procedure of opening an accounts in banks in some states for example in Ukrainian Banks it is difficult to open a business account and sometimes an application can be refused without explanation. With Bitcoin a business needs approximately 5 minutes to create Bitcoin wallet and immediately starts to use it without any questions and commissions. An additional benefit is its lower operating costs unlike physical cash there is no need to pay commission and fees to banks and other organisations. The commission fees in this system is lower than any other. It amounts to 0.1 % of transaction amount, with the operation interest changes going to the Bitcoin miner's wallets. Bitcoin ability to send money anywhere and to anyone in a matter of minutes gives it an upper hand than banks.

2.2.1 Regulatory Framework for Cryptocurrency

Among the many concerns in the dynamics of cryptocurrencies and blockchain ecosystems is the level of acceptance, the decentralised virtual currencies operating on distributed ledgers transactions are subjected to a great deal of speculation and the risk which calls for a direct enforcement of regulations in there ecosystem (Holding, 2019). According to Holding (2019), the steady increasing dissolute events in crypto world points in the direction for a call for sanity, however a few jurisdictions have taken pro-active steps in honing the potentials of the cryptocurrency markets in subjections to acceptable standards of operation. Herein the study will take a comprehensive look at the various burgeoning regulatory regime in a bid to create a suitable regulatory framework for the Zimbabwean Economy.

In Argentina under the National constitution of the country, the only authority responsible for issuing legal currency is the Central Bank. Bitcoin is not considered as a legal currency since it is not issued by the Central Bank but is viewed as money since they are not mandatory means of cancelling debts or obligations (Lauonia, 2018). Although the Bitcoins are not registered they are being used in the country and it is considered as a good or thing under the Civil Code and transaction with the Bitcoins is governed by the rules of sale of goods under the civil code. They have amended the Income tax laws that the profit derived from the sale of digital currency would be considered as income and taxed as such (Chomczyk, 2018).

Canada allows the use of cryptocurrencies, including Bitcoin. According to a Financial Consumer Agency of Canada webpage on digital currencies, "you can use digital currencies to buy goods and services on the internet and in stores that accept digital currencies, and you can also buy or sell digital currency on open exchanges called digital or cryptocurrency exchanges

(Financial Consumer Agency of Canada, 2018). However cryptocurrencies are not considered as a legal tender, but tax laws and rules apply to digital currencies such as the Income Tax. The Canada Revenue Agency has characterized cryptocurrency as a commodity and not government issued currency and its use is treated as barter transactions. Canada has amended its Proceeds of Crime and Terrorism Act to treat virtual currencies, including Bitcoin as a money service business for the purpose of anti-money laundering law (Ahmad, 2014). In 2017, Canada Securities Administrators published a staff notice on cryptocurrency offering which outlines how securities law requirements may apply to initial coin offerings and in 2018 approved the country first blockchain fund (Clara O' Hara, 2018).

The Switzerland government is trying to establish itself as a hub for cryptocurrency and Fintech start-ups. On 2 November 2017, the Commercial Register office in Canton Zug started to accept Bitcoin and Ether as a payment of administrative cost (Kanton Zug, 2017). Furthermore, the Commercial Registers accepts cryptocurrency as a contribution in kind for purposes of forming company whilst the municipal services (register registration) of up to CHF (about US\$210) can be paid with crypto (Kanton Zug, 2017). On February 16, 2018 the Swiss Financial Market Supervisory Authority published guidelines on the regulatory treatment of ICOs. In an ICO, investors receive blockchain-based coins or tokens in exchange for the funds they transfer which are then stored either on a blockchain specifically created for the ICO or on a pre-existing blockchain. In Switzerland, the individual cantons, the Swiss states, are obligated to levy income tax and wealth tax on the total property of taxpayers that are resident in their canton. Tax rates vary between the individual cantons. Cryptocurrencies are treated like foreign currencies for tax purposes and are subject to wealth tax.

In Mexico the Law to regulate Financial Technology Companies was enacted in 2018, includes a chapter on the operations of virtual assets. The chapter defines virtual assets as representations of value electronically registered and utilized by the public as a means of payment for all types of legal transactions, which may only be transferred electronically. In addition, Mexico has enacted a law extending the application of its laws regarding money laundering to virtual assets, thereby requiring financial institutions that provide services relating to such assets to report transaction exceeding certain amounts. The Mexico Central Bank granted the broad power under the law to regulate virtual assets to operate and authorizing financial companies to perform transactions with virtual assets.

In Venezuela under the Decree 3196 of December 2017, the government authorized to create its own currency, the petro, which would be physically backed by Venezuelas barrels of oil. One Petro would be backed by purchase-sale contract for one barrel of Venezuelan oil as quoted in the OPEC Reference Basket as well as other commodities, including gold, diamond, coltan, and gas. The Decree 3196 state that the holder of petro may be able to exchange the market value of the cyptoasset for the equivalent in another cryptocurrency or in Bolivares (local currency) at the exchange rate published by a national cyptoasset exchange house. The decree 3196 made ICO made through an action system or direct assignment by the Superintendence of Cyptoassets.

The German Federal Financial Supervisory Authority qualifies virtual currencies/ cryptocurrencies as units of account and therefore financial instruments (Banking Act, 2014). Undertaking and person that arrange the acquisition of tokens, sell or purchase tokens on commercial basis, or carry out principal broking services in tokens via online trading platforms, among others are generally required to obtain authorization from BaFin Advance. The Germany BaFin published information on the regulatory assessment of ICOs and the tokens, coins and cryptocurrencies they are based on. It stated that firms involved in ICOs need to assess on a case-by-case basis whether the ICOs qualify as financial instruments (transferable securities, units in collective investment undertakings or investments) or as securities and therefore trigger the need to comply with relevant legislation. In 2018 the Ministry of Finance published guidance on value-added-tax treatment of cryptocurrencies and stated that Bitcoin can be qualified as a virtual currency or digital money.

Most of countries in Africa has been issuing notices that ban the use of cryptocurrency, countries such as South Africa in December 2014 the South African Reserve Bank (SARB), the central banking institution whose responsibilities include formulating and implementing monetary policy and issuing banknotes and coins in the country, issued a position paper on virtual currencies. The position paper expressly stated that only the SARB may issue legal tender and that decentralized convertible virtual currencies (DCVCs), including bitcoin and litecoin, are not legal tender in South Africa. This means that “any merchant or beneficiary may refuse VCs as a means of payment (SARB, 2014). In August 2017 the Central Bank of Swaziland issued a press release regarding virtual currencies. The Bank noted that cryptocurrency does not enjoy legal-tender status in Swaziland. The Bank also noted that Bitcoin, a form of cryptocurrency currently being marketed and traded in Swaziland, is currently not regulated and this makes holding or trading the currency risky (CBS, 2017). In February 2018, the Zambia Securities

and Exchange Commission issued a notice on cryptocurrencies and other digital products. The Commission urged “any individuals or entities that are currently investing in or intend to invest in cryptocurrencies and related products/assets to exercise restraint and caution as they do so as the products/assets are largely unregulated and not subject to the jurisdiction of the Commission.” While it did not ban their operation, the Commission cautioned platforms that facilitate cryptocurrency transactions “to ensure that they are not in any way abrogating any part of the [Securities] Act and that those that meet the description of securities in accordance with the Act are registered with the Commission.

In a December 2017 statement the Reserve Bank of Zimbabwe noted that virtual currencies are not legal tender in Zimbabwe and remain unregulated. It issued the a warning to the public regarding the risks presented with the use of cryptocurrencies and those who participates in any transaction involving virtual currencies, does so at own risk and will not have legal protection from, or recourse against, any regulatory authority (RBZ, 2017). Therefore the study calls upon the adoption of the cryptocurrency and like other developed countries develop a regulatory framework that allows the operation of cryptocurrencies in the economy. The study will suggest a regulatory framework that best fits the Zimbabwean economy.

2.2.2 Cryptocurrency, liquidity and cost of international remittances

2.2.2.1 Liquidity Crisis

According to Walters (2020) a liquidity crisis is a financial situation characterised by a lack of cash or easily-convertible-to cash assets on hand across economies or financial institutions. In Zimbabwe the economy has been faced with a series of liquidity crisis since 2009 prior the dollarization period, were there was debilitating effect of hyperinflation, in 2013 the liquidity crisis emanated again after the dollarization period since the country was relying to a larger extend on export earnings and diaspora remittance as its own currency. In 2016 the liquidity crisis emanated again as the practice of money changing started, caused by businesses trying to find ways of coping with the crunch in the economy. The situation started again in 2019, when the country did not have sufficient liquidity to service its productive sector, since the country heavily relies on exports, they was need for more foreign currency which was scarce in economy (Southern Eye, 2019). Does cryptocurrency solve the liquidity crisis in the Zimbabwean economy? This therefore is the aim of the study.

In such as Canada and Germany were cryptocurrencies have been recognised as virtual currencies or digital currencies, they are being accepted as payments and settlement of debts

(World Press, 2012). Other countries such as Britain, they have allowed for Bitcoin ATM which the number of the ATMs has sprouted around Europe. Hundreds of merchants around the world are accepting cryptocurrency payments. Therefore an economy like Zimbabwe which have liquidity concerns may have to reconsider the use of blockchain technology in complementary with other fiat currencies.

According to Kampl (2014), cryptocurrencies usage in the economy have the ability to stop inflation that has been dominate in the Zimbabwe for many years. The maximum of coins is strictly limited to a certain number, there are neither political forces nor corporations that are able to change that order, and thus there is no possibility for the development of inflation in the system. Cryptocurrencies have unlimited amounts of transaction's that is, each of the wallet holders can pay to anyone, anywhere and any amount. The transaction can be controlled or prevented thus a user can make transfers anywhere in the world wherever another Bitcoin wallet is located. That it offer no transaction boundaries. The payments made in the system are impossible to cancel, the coins cannot be faked, copied or spent twice thus these capabilities guarantee the integrity of the entire system and allowing online shops, resources and companies to accept Bitcoin globally (Jonker, 2017).

According to Chitokwindo *et al.*(2014), the levels of financial inclusion in Zimbabwe are lower in rural areas despite 65 percent of the population are living in rural areas. From his survey, they had found out that most of the rural populace did not have a bank account because banks in Zimbabwe require either an income is processed in the account nor a minimal amount that can be left in the account for serving the account. Most banks are unavailable in rural communities which because a hassle to register an account (Chitokwindo *et al.*, 2014). With cryptocurrency it is easier to register and to use within 5 minutes after opening it. An additional benefit is its lower operating costs unlike physical cash there is no need to pay commission and fees to banks and other organisations. The commission fees in this system is lower than any other. It amounts to 0.1 % of transaction amount, with the operation interest changes going to the Bitcoin miner's wallets. Bitcoin ability to send money anywhere and to anyone in a matter of minutes gives it an upper hand than banks (Fahmy, 2018)

2.2.2.2 Cost of international remittances

The Zimbabwean economy has been backed by international remittances since independence, but however these personal remittance has increased from the year 2009 due to more migration of people to developed countries an indicate search of greener pasture (World Bank, 2018).

The value of personal remittances received in Zimbabwe was \$ 1 856 035 000 as at 2018, and this was a peak reached over 41 years an indication that the economy realise mostly on international remittances. According to the World Bank (2018), the total remittance totalled \$575 billion of which \$429 billion went to developing countries involving some 232 million migrants in which 10%, \$49 million constituted the cost of International remittances.

In many cases, the cost to consumers of these remittance transactions is expensive relative to the often low incomes of migrant workers, the amounts sent, and the income of remittance recipients. Therefore, any reduction in remittance transfer price would result in more money remaining in the pockets of migrants and their families, and would have a significant effect on the income levels of remittance families. Indeed, if the cost of sending remittances could be reduced by significantly relative to the value sent, remittance recipients in developing countries would receive over \$16 billion dollars more each year than they do now. This added income could then provide remittance recipients more opportunity for consumption, savings, and investment in local economies (World Bank, 2018). This therefore calls for cryptocurrency in developing nations.

According to World Bank (2018), remittance prices are high for many reasons, including underdeveloped financial infrastructure in some countries, limited competition, regulatory obstacles, and lack of access to the banking sector by remittance senders and/or receivers, and difficulties for migrants to obtain the necessary identification documentation to enter the financial mainstream. Cryptocurrency has become the answer to this problem, it offers a new remittance solution that is significantly faster, cheaper and easier than the traditional model. Cryptocurrencies lower transaction costs and speedy cross border payments. The offer lower operating costs unlike physical cash, there is no need to pay commission and fees. The commission fees in this system is lower than any other. It amounts to 0.1 % of transaction amount as compared to a 7.09% of remittance companies, with the operation interest changes going to the Bitcoin miner's wallets. Bitcoin ability to send money anywhere and to anyone in a matter of minutes gives it an upper hand than banks (Fahmy, 2018). The Internet enables Blockchain, granting universal access, which can be used as a means of financial integration, making payments faster and cryptocurrency-based money transfers are also more secure.

2.2.3 Opportunities and threats of cryptocurrencies in the economy

2.2.3.1 Opportunities

Cryptocurrency is in a unique position as a forerunner in a possibly transformative technology to long standing financial systems. By its very nature, it is able to fill gaps in current financial technologies and be able to help solve traditional banking problems by being a peer-to-peer system. Napster, another peer-to-peer system, transformed the music industry by cutting out the middle man (Kelly, 2014). Transformative technologies start by solving a specific problem in an industry. For instance, cryptocurrencies are poised to help remediate the problems related to unbanked consumers. Significant portions of the population in developing countries are unbanked. In Latin America, 60% of 600 million inhabitants have no access to bank accounts (Magro, 2016). Bitcoin's technology allows for individuals to exchange currency without needing a third trusted party, like a bank, to oversee the transaction. All that is needed to use Bitcoin is a mobile phone, which 70% of Latin Americans do have access to (Magro, 2016). Due to Bitcoin's ad-hoc networking capability, two users can trade Bitcoin with each other by scanning QR codes displayed on their phones printed out by the application. This is a truly unique solution to a problem that has existed for many years for some people. This would invariably increase as the user base grows, so the demand for better cryptocurrency network and applications will come to the forefront.

Businesses are beginning to see the value in using cryptocurrencies for international transactions, especially when transactions need to occur quickly in response to an emergency. Cryptocurrencies are solely positioned to solve this problem thanks to the speed and ease of transaction in the peer-to-peer system. Money can be wired internationally, but typically arriving days after being sent and not for the full amount (Team, 2016). The transaction can be hit with any number of unexplained fees as it crosses borders, making it difficult to send the correct amount to another business. A good example of this type of emergency need is an online company who is suffering from a denial-of-service attack and is looking to get immediate protection from a network security company (Team, 2016). In this scenario, speed of transaction is of the essence, for every minute that the company's website is down, profits are being lost. Cryptocurrency has a major advantage over traditional currencies thanks to its agility in making fast peer-to-peer transactions, especially in international business-to business scenarios.

Bitcoin has developed a questionable reputation through recent events. Stories like Silk Road can portray a negative image of digital currency in general, not just Bitcoin. Silk Road was an online marketplace buried in the dark-net, which allowed thousands of drug dealers and nearly a million customers to make illegal drug deals. Bitcoin was their primary means of transaction, due to the lack of government tracking and semi-anonymity. It ran from 2011 to 2013, and racked up nearly one billion USD in sales (Bearman, 2015). People want criminals to have justice meted against them, so the semi-anonymity attribute of bitcoin seems negative to law abiding citizens. Without positive marketing towards the value of semi-anonymity for normal users, the general user base will think that cryptocurrencies are only used by criminals.

International laws regarding taxation have been passed recently, creating validity for cryptocurrency as a mainstream device. Laws regarding the taxation of cryptocurrencies are required before digital currency could be considered a truly valid form of transactions. Towards the end of 2015, the European Court of Justice announced that it viewed bitcoin transactions as exempt from value-added tax (Hileman, 2016). Steps like this will significantly increase cryptocurrency flow. Some users would refuse to use currency without knowing how it would affect their tax statements positively, regardless of what negative light in which they are viewed.

One of Bitcoins largest opportunities is that it can also act as a sort of commodity, similar to gold. The value of gold can spike considerably whenever an event threatens the balance of the global market, as we have seen with the Brexit vote. The precious metal saw an increase in value to a two-year high as investors became uncertain as to how the markets would react to the vote, using it as a safe haven (Reuters, 2016). The commodity market is a widely accepted form of trade worldwide, and cryptocurrency has seemingly begun to mimic the characteristics of gold. Gold has been a long standing holder of value, and that is based on the universal acceptance and trust of its value. Cryptocurrencies could potentially become a big player in the commodity market. They have a unique attribute of being purchased through a direct online mechanism, which creates easy entry for buyers. If bitcoin continues to be a valid refuge for inflating currencies, it will gain validity to investors and push deeper into becoming more mainstream.

Bitcoin has strength by design to make it a viable currency that has elevated it in status over the years, more notably the fixed limit of bitcoin that will exist. Bitcoin will be mined with diminishing returns every four years until the maximum number of bitcoins are reached: a total

of 21 million (King, 2013). This aspect of Bitcoin is important for its value. Due to the limited amount of bitcoins, it will never become inflated from an overabundance of bitcoins. Also, bitcoin and other cryptocurrencies are generally regarded as being protected from inflation originating from national government changes or restrictions (Magro *et al.*, 2016). This creates a “safe haven” for investors to put their wealth into, as it generally does not lose value based on inflation. Bitcoin is quickly showing its strength as a refuge against inflating national currencies. However, as is the case with most commodities, the price can fluctuate wildly based on many other external factors. The combination of demand for a safe haven option and its price volatility helped Bitcoin to become the best performing currency of 2015 using the US Dollar Index (Desjardins, 2016). This means that Bitcoin was the highest valued currency in the entire world at the end of last year. This is no small feat in a global economy with powerhouses like China and the United States running the landscape.

2.2.3.2 Threats

Bitcoin has quite a few hurdles to clear for user acceptance to become widespread. The value fluctuations that plague cryptocurrencies puts doubt in users, as well as investors. Ultimately a limiting factor in cryptocurrency is general acceptance (PwC, 2019). Value fluctuations reduce trust that a consumer’s value would be retained on a day to day basis, limiting faith in the currencies overall worth. In a survey performed by PwC, 83% of those surveyed had little to no familiarity of Bitcoin (PwC, 2019). The lack of central ownership of cryptocurrencies means that any attempt to remediate this marketing problem using advertisements could theoretically help the investing company’s competition. This is not an ideal situation for a marketing plan.

According to Pwc (2019), cryptocurrencies have also seen fraud and theft, generally due to faulty system setups by exchange companies. These hacks generally make the news, and can easily convince the layman that they are unsafe locations to put their money. There is also a large gap in laws that cover the use of cryptocurrency. As long as cryptocurrencies remain in an area not generally covered by law, user acceptance will be limited. User’s need to trust that any transactions using cryptocurrencies are legal and binding. Markets and governments are slow to react to the new technology. Ultimately, all of these factors limit consumer’s trust in Bitcoin and cryptocurrency.

This lack of trust leads to issues with investors as well. The dead pool of failed start-ups has increased to 24, mostly citing security as the main reason for closure (Hileman, 2016). This metric could be considered a watermark for future investors to consider before investing in

Bitcoin. The Mt Gox and DAO hack shows how inattentive organization can not only lose millions of dollars" worth of digital currency, but can drop the value significantly. New start-ups now know that a haphazard and unplanned launch is ill-advised at best, and new market entry will be limited. This could ultimately hurt Bitcoin, as development of better software is important to improve security and user acceptance. As obvious of a concern as it may seem, security implementation and fixes are both generally slow to adapt for any new technology. Even the DAO hack exploit was documented as a potential problem weeks before the attack (Price, 2016). One of the issues with security is that the decentralized nature prevents a unified effort to completely secure every server that runs the code. A unified front in the realm of cryptocurrency may need to rise before the peer-to-peer network would become truly secured.

Another serious threat to cryptocurrency is the maze of regulations that would need to be traversed before mainstream user acceptance. Most of the government not even classify what type of asset Bitcoin is, which will prevent most market participants from adopting cryptocurrency-based business models (PwC, 2015). Cryptocurrency could be labelled as either a security, capital asset, commodity, or a currency, and each would have a different effect on how Bitcoin is adopted. International views of Bitcoin vary by country, but seems to be viewed positively based on Bitpay"s assessment of transactions. In Europe, transactions have reached an all-time high at 102,221 per quarter (Patterson, 2015), which may be the cause regulations being passed regarding Bitcoin and cryptocurrency. Bitcoin transaction have become exempt from value added tax by the European Court of Justice, effectively recognizing it as a legitimate means of payment in Europe (Perez, 2015). This simply means that Bitcoin transaction will not be taxed in Europe. While great news for European Bitcoin users, other major markets are still missing crucial legislation regarding Bitcoin taxation.

2.3 Theoretical Framework

This section explore theories that can back-up the adoption of the new technologies in this fourth world industrial revolution. The study adopted the Innovation diffusion theory by Rogers (1962) and the Disruptive innovation by Christensen (1971). These theories will provide the theoretical framework of the study.

2.3.1 The Diffusion of Innovation Theory

The diffusion innovation theory was developed by Rogers in 1962 and it explains how overtime an idea or product gains momentum and diffuses or spreads through a specific population or social system. The end result of this diffusion is that people in the social system will adopt a

new product, new innovation, brand or product. The adoption means that an organization/country will have to do something different than what they had previously done, that is a change in adoption, product offering or performing a new task. The key to adoption is that the organization or countries must perceive the idea, implement the new innovation, so as to make the diffusion possible.

According to the diffusion theory, adoption of a new idea, brand or product (innovation) does not happen simultaneously in a social system; rather it is a process whereby some organizations or people are quicker to adopt the innovation than others. The early adopters of innovation have been found to be blue chip organizations or countries who have different characteristics those who have to adopt the innovation at a later stage. When promoting innovation to a target population it is important to understand the characteristics of the target population that will help or hinder adoption of the innovation. Rogers (1962, p13) adopted five categories of organizations or people and the time that they would adopt the new technology and the majority of the general population tends to fall into the middle categories, however it is still necessary to understand the characteristics of the target population. When promoting an innovation, there are different strategies as well that can be used to appeal to different adopters. The category is as follows:

Innovators: these are the countries that want to be the first to try innovation. They are venturesome and interested to try new ideas. These countries are very willing to take risks and are often the first to develop new ideas. Very little, if anything needs to be done to appeal to this population. Relating to this research examples of these countries are Japan, Russia, Switzerland, Serbia and Tajistan.

Early adopter: these are the countries who represent opinion leaders, they enjoy leadership roles and embrace change opportunities. They are already aware of the need to change and so are very comfortable in adopting the new ideas. The strategy to appeal to this group of the population includes how-to manuals and information sheets on the implementation; this group does not need information to convince them to change.

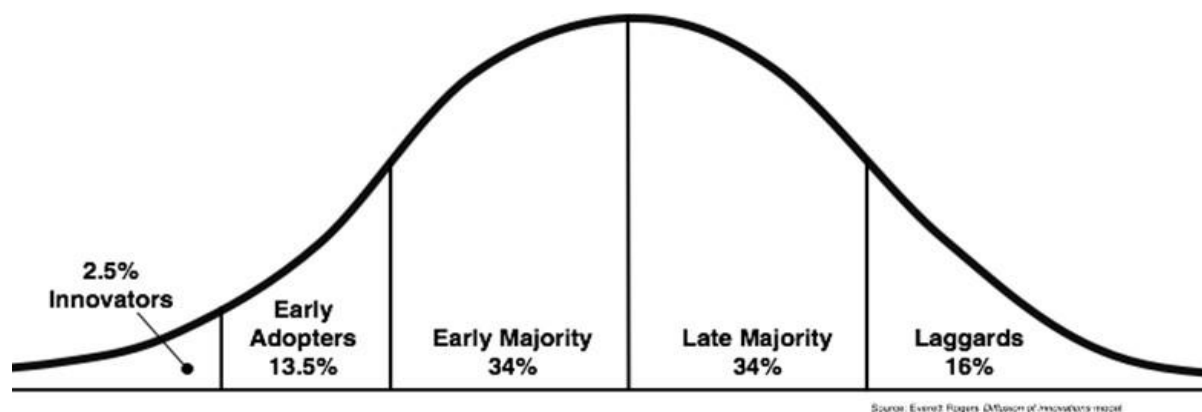
Early majority: these countries that are rarely leaders, but they do adopt new ideas before the average country. Typically they need to see evidence that the innovation works before they are willing to adopt it. Strategies to capture the early adopters include success stories and evidence of the effectiveness of innovation.

Late majority: these are countries that are sceptical of the change, and will adopt an innovation after it has been tried by the majority. The strategy to capture this group of counties includes the information on how many other countries have tried the innovation and of how many have adopted it successfully.

Laggards: these are the countries that are bound by their tradition and are very conservative. They are very sceptical to change and are the hardest group to bring on board. Strategies to appeal to this population include statistics, fear appeals and pressure from organizations who have already adopted the technology. These represent most of the African countries.

Fig 2.3 shows the diffusion innovation theory stages.

Fig 2.3 The Diffusion Innovation Theory



Source: Adapted from Rogers (1962)

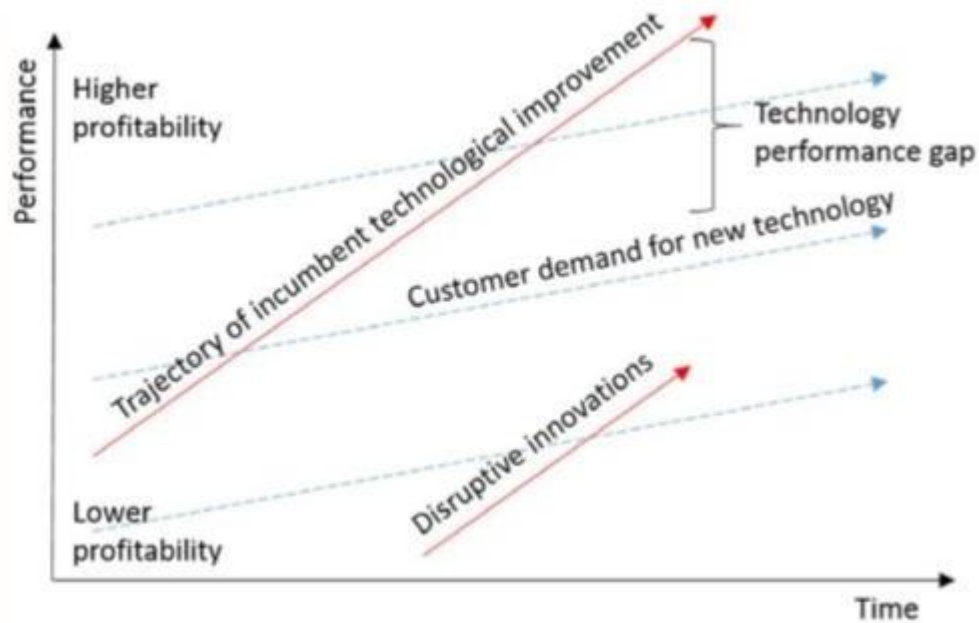
The diffusion of innovation theory is applicable in the study as it shows the phrase at which the countries are willing to adopt to the technological innovations. In terms of cryptocurrency adoption, these phrase are being evidenced for example Japan who have been the innovators of blockchain technology was not reluctant in accepting the proposal of cryptocurrency after the submission of a working paper in 2008 by Satoshi Nakamoto. Taking evidence from the European and Central Asian countries constitute more of the early adopters and early majority. Examples of the countries are Russia, Switzerland, Estonia, Georgia, Venezuela and Ukraine. The late majority are mostly countries in African continent for example South Africa has been making way in creating a regulatory framework for the use of blockchain technology, in Kenya, the government are now taxing the new digital currencies under the Finance Bill and in Nigeria through the Nigeria Securities and Exchange Commission are setting up a framework for the

regulation of Virtual Financial Assets and their Exchanges. Currently the bulk of the laggards are countries again in Africa such as in Burundi, Morocco, Egypt and Zimbabwe who have put a complete barn to the trade and use of cryptocurrencies.

2.3.2 Disruptive Innovation Theory

The disruptive innovation is an innovation that creates a new market and value network and eventually disrupts an existing market value network, displacing established market leading firms, products and alliances (Airini *et al.*, 2017). The theory of disruptive innovation was first propounded by Christensen in 1997, where the first examined the drive industry. The results of this multi-method study indicated that, when an innovation emerged that improved performance on dimensions that customers historically valued (e.g., the capacity and recording density of disk drives), incumbents tended to lead commercialization and to maintain their market position. However, when an innovation emerged that did not improve performance along this customer-preference trajectory but introduced a unique constellation of attributes (e.g., small, lightweight, rugged), new entrants led development while incumbents languished or failed. This pattern was observed consistently across multiple technological generations and product life cycles (Christensen *et al.*, 1998; Rosenbloom and Christensen, 1994). Fig 2.4 shows the model of disruptive innovation.

Fig 2.4 Model of Disruptive Innovation



Source: Adapted from Christensen (1997: p 56)

From his study of the disk-drive industry, Christensen (1997) induced an account of disruptive innovation that consisted of three principal components. First, in many industries, the pace of technological progress outstrips customers' demand for higher-performing technologies. As a result, incumbents can over-serve the market by producing more advanced, feature-rich products than customers need; doing so leaves a gap at the bottom of the market between customers' needs and the performance provided by firms a gap that provides an opening for entrants (Fig 2.3). Second, for firms, a strategically crucial distinction between different types of innovation in technology or in business models can emerge in an industry. Most are sustaining innovations, which improve products and services along dimensions of performance that mainstream customers care about and that markets have historically valued; such innovations enable incumbents to sell more products to their best existing customers at higher margins and higher profitability.

The rarer type is a disruptive innovation. When initially introduced, disruptive innovations are inferior to incumbent products on accepted performance dimensions, but they offer a novel mix of attributes that appeal to fringe customer groups, notably those near the bottom of the market. They may be, for instance, smaller, cheaper, more accessible, or more convenient. Finally, the third component of Christensen's model was that existing customers and established profit models constrain established firms' investments in new innovations; thus, investments unattractive to incumbents may be attractive to entrants who lack many (or any) customers and

enjoy fewer competing investment opportunities. Consequently, incumbents are typically unmotivated to develop disruptive innovations that promise lower margins, target smaller markets, and introduce inferior products and services that their existing customers cannot use.

According to Christensen (1997), argues that disruptive innovations can hurt successful, well-managed companies, O'Ryan (2016) countered that constructive integration of existing, new, and forward-thinking innovation could improve the economic benefits of these same well-managed countries, once decision-making authorities has understood the systemic benefits as a whole. Christensen distinguishes between low-end disruption, which targets customers who do not need the full performance valued by customers at the high end of the market, and new-market disruption, which targets customers who have needs that were previously unserved by existing incumbents. Low-end disruption occurs when the rate at which products improve exceeds the rate at which customers can adopt the new performance. Others, at some point the performance of the product overshoots the needs of certain customer segments. At this point, a disruptive technology may enter the market and provide a product that has lower performance than the incumbent but that exceeds the requirements of certain segments, otherwise gaining a foothold in the market.

2.4 Empirical Literature Review

This section of the literature presents past studies that were conducted on the contracts of the study in a bid to come up with a gap that the present study seeks to cover. Below is the empirical literature review of the study:

In a study by Naboulsi and Nuebert (2018) on the impact of digital currencies on economic development in Kenya. The paper aimed at identifying and investigating how digital currency has had an impact on the Kenyan economy. The study aimed to show how using Bitcoin can reduce the cost of international funds transfers and reduce remittance fees subjected from other transfer providers. The findings of the study suggest that there are concerns about Bitcoins use including its price volatility and unregulated legal framework put in place on the use of digital currencies. The outcomes suggest that the use of BitPesa and Bitcoin technology had some impact on users understanding and adopting to the use of this technology and how it works. The results prove that using BitPesa to transfer money has reduced the cost of transfer on users.

In study by Moodley (2019), on Examination of Blockchain Technology and Venezuela's Sovereign Based Cryptocurrency Including the Effects of Implementing a Sovereign Based

Cryptocurrency in Developing Countries. The study sought to examine cryptocurrencies and blockchain technology by looking at Venezuela's Petro as the model to assess whether such a development would be beneficial to a developing country such as South Africa. In addition, any risks that are attached to this invention will be examined. This research paper recommends that the developing countries consider implementing blockchain technology in various institutions as this technology has many uses and advantages such as assisting with the voting process, as well as minting citizen's information and the South African Reserve Bank to research the area of blockchain technology in the banking system as this would decrease interbank transactional costs thereby allowing more transactions to be completed in a shorter amount of time.

In a related study by Mahomed (2017) on understanding consumer adoption of cryptocurrencies. The study sought to build an understanding the drivers of consumer adoption of the technology. The findings indicate that facilitating conditions have the highest explanatory effect on actual usage ahead of behavioural intention to use cryptocurrency. Behavioural intention was predicted most strongly by hedonic motivation, followed by perceived trust, and social influence. Interestingly, effort expectancy and performance expectancy were found to be non-significant, contrary to much of the studies in related fields.

In a study by Mazikana (2018) on the impact of cryptocurrencies in Zimbabwe an analysis of Bitcoins. The study sought to find out if Bitcoins can solve the liquidity crunch in Zimbabwe and assess the factors that influence the adoption of Bitcoins by banks in the country. The study made use of primary data which was collected through the use of questionnaire with respondents selected from a sample of banks such as CBZ, Steward Bank and Cabs. The study adopted a sample size of 50. The study concluded that there are many meaning attached to cryptocurrency, it is beneficial to adopt cryptocurrency and there are many challenges faced cryptocurrency. The study recommended that banks in Zimbabwe have to adopt cryptocurrency as a way to enhance its payment facilities.

2.5 Research Gap

Virtual currencies have become the new normal in the financial space particularly the crypto currencies. Although much has been written about the significance of crypto-currencies the world over much research has not been done to establish the impact of cryptocurrency in Zimbabwe because of its ban in 2018. The literature on crypto currencies has proved that there

is a research gap on the impact of the adoption of crypto currencies and its effect on the economy. The effects of crypto currencies and the feasibility of it through the use by banks and individual is currently unknown. It is the aim of the study to close the knowledge gap, by showing how the emergency of crypto often referred to as disruptive technology create a bridge between information technology and finance because changing the way some institution works. The study will explore the feasibility of the crypto currency in the Zimbabwe economy, regulatory framework that needs to be put in place for its smooth operation, the effect of cryptocurrency to cure economic problems such as financial issues like liquidity problems and the opportunities and threats that come after the adoption of cryptocurrency in the economy.

2.6 Summary

This chapter provided a critical review of the literature on the impact of the adoption of cryptocurrencies in Zimbabwe and its effect on the economy. The study variable that is cryptocurrency and its operations were well defined. The study adopted the Innovation of Diffusion Theory and Disruptive Innovation Theory as the theoretical framework which was well described and analysed. The last part of the literature identified the gaps that the present study seeks to cover. The following chapter looks at the research methodology to be employed in this study.

CHAPTER III

RESEARCH METHODOLOGY

3.0 Introduction

This chapter presents the research methodology undertaken by the study. It outlines the research design, philosophical paradigms, describes the data collection instrument, defines and describes the target population, sampling, data sources, data analysis technique, research validity and reliability, ethical considerations and limitations. The justification for any methodology used is important in this chapter. The chapter ends with a chapter summary.

3.1 Research Philosophy

A research philosophy is a belief about the way in which data about a certain phenomenon should be gathered, analysed and used. The study adopted a positivist research philosophy which was combined with a quantitative methodology in collecting data. According to Greener (2010) positivist philosophy entails a belief that the reality is stable and can be observed and described from an objective viewpoint without interfering with the phenomena being studied, thus the phenomenon should be repeatable. Positivism puts an emphasis on the scientific empirical methods designed to yield data and facts that are not influenced by human interpretation. Positivist research philosophy adopts an epistemology approach (what constitutes acceptable knowledge), and focuses on discovering observable and measurable facts and phenomena which can lead to meaningful results of the data collected.

The study also adopts a deductive approach, which is more suited to the positivist philosophy which allows for the formulation of a research hypothesis and formulation of a strategy to test the hypothesis (Snieder and Larner, 2016). The study made use of questionnaire, interviews and desktop research which is compatible with the deductive approach. The rationale for the research approach is that the main constructs of the study are measurable and have evolved from the body of theory.

The study adopted a positivist philosophy because the approach believes that the reality is stable and can be observed and described from an objective viewpoint that is without interfering with the phenomenon being studied which suits the study. The positivist approach content that the phenomenon should be repeatable which makes the study valid and reliable. The study made use of a single independent variable (cryptocurrencies), so as to form

relationships between the dependent variables which is compatible with a positivist research philosophy. The study predictions are also being made on the basis of the previously observed and explained realities and interrelationships of which the approach chosen is able to support the notion.

3.1.1 Research Approach

A quantitative deductive approach was used to test the effectiveness of cryptocurrency in the economy. Using data collected as described in the following sections, the researcher aimed to discover the major drivers of adoption among cryptocurrency users and its effect on the economy. The research is therefore explanatory in nature and the quantitative design follows a positivist approach used by the study (Sunders and Lewis, 2012).

3.1.2 Research Strategy

The research made use of interviews, questionnaire and desktop research as a way of collecting primary data and secondary data. Direct measurement of cryptocurrency users is difficult due to the pseudo-anonymous architecture of the technology (Bohr & Bashir, 2014), therefore the researcher did not include cryptocurrency user. The research used respondents from the RBZ, Ministry of Finance and cryptocurrency firms. The research instruments were pilot tested amongst a small group of cryptocurrency user online to ensure that the content is valid to testing of the above hypotheses.

3.1.3 Time Horizon

The time was determined by the Great Zimbabwe Banking and Finance department was less than a year for the academic submission deadlines. Thus since they were only six months to carry out the research, a cross sectional study was used.

3.2 Research Design

A research design is a conceptual blueprint within which a research is conducted, thus it is a preparation of the action plans, constituting the outline of the collection, measurement and analysis of data (Akhtar, 2016). There are seven research designs that the researcher may make use of when carrying out a research namely explanatory, experimental study, entomology, case study design, action research, archival research and grounded theory design (Williams, 2007). The choice of a research design is influenced by many factors such as the methodology applied, approach proposed, philosophy ascribed and the availability of time and other resources needed to conduct the study efficiently.

This study adopted a quantitative research design. Quantitative research design entails the methods of measurement and the statistical, mathematical, or numerical analysis of data collected through questionnaires and surveys or by manipulating pre-existing statistical data using computation techniques. The researcher's choice of research design follows a number of aspects. Quantitative design approach allows the researcher to quantify data easily. The research design is compatible with research instruments like questionnaires and interviews which have the ability to collect huge amounts of data at a shorter period of time, which will make data easier to quantify. The research design is also compatible with a positivism philosophy which makes use of a deductive approach, hence quantitative research design has been considered to be the best to use in this study.

3.3 Research Design

According to Elmabruk et al. (2018), a research design refers to the overall strategy that the researcher chooses to integrate the different components of the study in a coherent and logical way thereby ensuring that the study effectively address the research problems; it constitutes the blueprint for the collection, measurement and analysis of data. The study adopted an explanatory research design. An explanatory research design focuses on gaining new insights and familiarity for later investigation or undertaking when the problems are in the preliminary stage of investigation (Boru, 2018). It focuses on explaining the research objective of the study in a detailed manner, generates new ideas and assumption development of theories or hypotheses and provides a well-grounded picture of the situation being developed. The study adopted an explanatory research design because it allows the study to provide a deep insight into the specific subject which gives birth to more subjects and provides more opportunities for the researcher to study new things and question new perspectives. It creates a cycle and critical thinking of the study which creates more questions of which those questions lead to more ways for the researcher to have different perspectives related to the study.

3.4 Target Populations

The target population of the study was made up of three different types of respondents that is employees from the cryptocurrency firms, the Ministry of Finance and the Reserve Bank of Zimbabwe. So far in Zimbabwe there are 2 cryptocurrency firms and office in Harare and Bulawayo for the Ministry of Finance And RBZ. The cryptocurrency firms gave the researcher information pertaining to; how they are operating without a regulatory framework, what would be there proposed regulatory framework and what they need for the government to do to

support cryptocurrency activities. The Ministry of Finance is the governing authority of all financial activities in the country, thus information like the re-consideration of cryptocurrency ban, also their thoughts in reconsidering research on cryptocurrency and who do they make statutory instruments that support cryptocurrency operations is obtained. The Reserve Banks of Zimbabwe as a regulator of fiscal and monetary policies its information would help in the discussion of the way forward for cryptocurrencies to be able to operate in the economy and how it can be controlled. Table 3.1 shows the proposed target population for the study.

Table 3.2 Target Population of the Study

City	Reserve Bank Representatives	Ministry Of Finance Representatives	Cryptocurrency Firms Representatives	Target Population
Harare	100	80	30	210
Bulawayo	100	50	20	170
Total				380

3.5 Sample Size

According to Malone (2016), sample size determination entails the act of choosing the number of observations that would be included in a statistical sample, thus a study sample size is an important feature in any empirical study in which its goal is to make inferences about a population from a sample. A large sample produces information that is adequate for the research to produce accurate results (Olivera and Martins, 2011). Table 3.2 shows the sample size of the study.

Table 3.2 Sample Size

City	Target Population	Sample at 85%
Harare	210	170
Bulawayo	180	128
Total	380	323

Table 3.2 shows the sample size of the population, the study adopted 85% of the target population. A sample size of the study has to be sufficient enough to produce accurate results. According to Brick and Wood (2014), a sample size of 30% or more of the target population for an empirical study is sufficient enough to produce accurate results. The study adopted 75% of the target population, therefore basing on Brick and Wood (2014), the sample size adopted by the study is sufficient enough to justify the results of the study.

3.5 Sampling Techniques

Sampling is the selection of a subset or a statistical sample of individuals from within a statistical population to estimate the characteristics of the entire population (Hickman, 2013). The study made use of stratified random sampling in the selection of the respondents. At first the researcher made use of stratified sampling, in the selection of respondents into stratas based on the similarity of their characteristics. Stratified sampling entails the subdivision of the population into separate groups (stratas) based on similar characteristics that the respondents have. The study had chosen the respondents based on their characteristics that is the Reserve Bank of Zimbabwe employees (the regulator), Ministry of Finance employees (the governing authority) and the cryptocurrency firms.

After the study employed stratified sampling, a simple random sampling was adopted. Simple random sampling entails the probability that every item in the population has an even chance and likelihood of being selected in the sample. Simple random used to select respondents that were chosen with stratified sampling based on their convenience to the researcher and the researchers' judgment that the respondents may provide relevant information for the study.

The study made use of stratified random sampling because it provides a greater precision that requires a small sample size. Stratified random sampling accurately reflects the population being studied because the researcher stratified the entire population before applying simple random sampling thus it ensures each subgroup within the population receives proper representation within the sample. As a result stratified random sampling provides a better coverage of the population since the researcher would have control over the subgroups to ensure all of them are represented in the sample.

3.6 Research Instruments

According to Malhotra (2016), research instruments are tools that are used to collect, analyse and measure collected data from a sampled population. Research instruments are measurement tools (questionnaire and interviews) designed to obtain data on a topic of interest from research subjects (Akhtae, 2016). The study made use of questionnaires and interviews in collecting primary data. A questionnaire is a preformulated set of written questionnaires which respondents record their answers within closely defined alternatives (closed ended questions) or in their own words (open ended questions) (Zull, 2016). According to Frances (2009), an

interview is a structured, semi-structured or unstructured conversation where an interviewer asks a question about a certain phenomenon and the interviewee provides answers.

The researcher made use of questionnaires to gather data with the RBZ and Ministry of Finance employees selected with stratified random sampling. The researcher made use of semi structured interviews to the 2 cryptocurrency firms which gave room for the respondents to explain further a certain phenomenon, give clarity and ask for further explanation of the questionnaires. The interview questions were made in a way that allowed the researcher to obtain adequate information from the interviewee. The researcher made use of closed ended questions which allowed the respondents to answer within closed defined alternatives related to the study. The questionnaire used a likert scale so that the respondents rank the extent they strongly agreed or strongly disagreed to questions.

The study adopted questionnaires because they provide a relatively cheaper, quick and efficient way of collecting large amounts of information from a large sample. The format that is used in the questionnaires is familiar to most respondents which saves on time, the information collected, is collected in a standardised way which makes it easier for data analysis. Interviews were chosen for the study because they collect rich data from a small number of people. They help to explain and have a better understanding of the certain phenomenon of which the research subject's opinion, reaction, behaviour, experiences on a certain question is shown.

3.7 Data Sources

Data sources are devices or tools that are used to measure data which was collected (Douglas, 2015). The researcher made use of both primary and secondary data sources. According to Ajayi (2017), primary data source is the one which is collected for the first time by the researcher while secondary data uses the data that is already available, which was produced by other scholars. The researcher made use of primary data sources through the use of information collected by questionnaires and interviews, on the other hand secondary data sources were used in the information that was collected from previous publications, journals, websites and articles. Both primary and secondary data was used by the researcher in data presentation, discussion and analysis.

The researcher made use of primary data because the information that is collected is particularly for the study with the main aim to find the answers of the research questions addressed in the research introduction. In that essence, the questions the research asked were

tailored to obtain information that is used for the study. Primary data is convenient to the researcher since it is collected by himself using interviews and questionnaires. Secondary data is readily available and inexpensive to obtain. Additionally, secondary data tends to have a large sample because the data collection is comprehensive and routine, that is more data is collected over a long period of time using different types of secondary data. This allows the researcher to detect changes in the literature overtime and find the research gap that the study may explore.

3.8 Data Collection Procedures

Data collection entails the process of gathering and measuring information variables in an established and systematic fashion that enables the researcher to answer stated research questions, test hypothesis and evaluate outcomes (Muhammad, 2016). In the study both primary and secondary was used to collect data. In data collection the researcher made use of a process that had 4 key steps, thus; by firstly determining what the information the research study wants to collect. At this stage, the researcher chose the details that the study wants to collect, that is the decision that the topics would cover, what to collect and how much data which needed. On the second step, the researcher set a timeframe for the data collection that is by formulating a plan for how the data would be collected, the researcher established a time frame for data collection. On the third step the researcher determined the data collection method, where on this step a chosen data collection method that makes up the core of the data gathering strategy. Lastly, the researcher collected the data once the plan was finalized. The researcher implemented the data collection strategy and started to collect data successfully.

In administering questionnaires, the researcher made use of the drop and pick approach whereby the researcher dropped the questionnaires to the cryptocurrency firms and picked them up after a day, when they had answered the questionnaire. Interview appointments were scheduled with the senior management through walk in and telephone appointments on different days depending on the availability of the employees.

3.9 Validity and Reliability

Validity and reliability of the research entails the concepts used to evaluate the quality of the research, they indicate how well a method technique or test measure is valid. Reliability entails the consistency of a measure to produce the same results if the research was to be repeated. It is important for the research to incorporate concepts of validity and reliability in the research.

Below entails the steps that the researcher may use to incorporate the concepts of validity and reliability of the research.

3.9.1 Validity

According to Kubai (2019), validity gives an explanation of how well the collected data covers the actual area of the research thus, it basically measures what is intended to be measured in the research. In order to ensure that the research incorporate the aspects of validity the researcher ensured that the information collected was valid by using two data collection instruments and a pre-test of the study before the actual study was employed. Firstly the researcher carried a pilot testing of the study to a small group of employees that possessed the same characteristics of the research subjects of the study. The researcher noted some corrections on the research instruments and adjustments were made so as to collect rich and accurate data for the study. The researcher made use of questionnaires and interviews due to the complementary effect that the instruments poses on each other, that is each of research instruments reduces the shortfall of the other when used separately. In addition, the researcher avoided the use of ambiguous or double barrelled questions that might compromise on the response; these simple and straight forwarded questions were used to ensure that all the respondents have a better understanding of the questions. After the researcher was satisfied that the data collection tools are effective, the questionnaires were administered to the sample bias free and in person assuring them of privacy of their responses. Necessary follow ups were made to receive the questionnaires and setting up of appointments with the required respondents.

3.9.2 Reliability

According to Mahojan (2017), reliability of the study entails the repeatability of the findings, that is, if the study was to be done for the second and third time, it would produce the same results, therefore if the research study yields the same results for the second or third time, it is said to be reliable. Reliability is all about the consistency of the research instruments to generate the same responses under the same conditions. For the research to incorporate the aspects of reliability, the researcher made use of a triangulation method. A triangulation method means the use of more than one method to collect data on a certain topic. The researcher made use of self-administered questionnaires and structured interviews. The two research instruments are complementary since the weaknesses of the questionnaires are reduced by interviews. Questionnaires were administered bias free, since the researcher first conducted a pilot study before the actual questionnaires were distributed. Interviews allowed respondents to explain

further areas that needed clarification and to answer questions that are in the minds of respondents that are not included in the questionnaires.

3.10 Ethical Consideration

According to Akaranga (2016), a study needs to be sensitive to the vulnerable respondents and respect the power inequalities by not placing the respondents at risk. To ensure that the research ethics were maintained, the researcher issued an informed consent that is fully informing the respondents about the research being conducted. The respondents were fully informed about the purpose of the research, who or what group is funding it and how the finding will be used for. The researcher ensured voluntary participation of the respondents by making the respondents participate free from any coercion. The respondents were assured of withdrawal of their participation at any time without negatively impacting on their involvement in the future projects and relationship with any of the researchers or research bodies. The researcher ensured that the researcher may not be harmed in any way that is either physically nor physiological which can be in the form of stress, pain, anxiety, diminishing self-esteem or invasion of privacy. The researcher also ensured confidentiality of the respondents by making sure that the identifying information is not available to, or accessed by anyone rather than the researcher. The researcher ensured anonymity of the researcher by making the respondents not to disclose their names and remain unknown. The researcher only assessed the components that were relevant to the research subject being conducted. The researcher assumed full responsibility for disclosing the methods used while conducting the research, limitations encountered and the positive benefits derived from the study to the finance sector, Great Zimbabwe University and other stakeholders including the whole world at large.

3.11 Data Presentation and Analysis

Malhotra (2016), argues that data presentation refers to the organization of data into tables and graphs or charts so that logical and statistical conclusions can be derived from collected measurements. Data analysis is a systematic approach to cutting down or condensing of accumulated data into usable quantities, it involves the summarizing of collected data outlining the patterns of results by using a statistical tool with the purpose of identifying useful information that the researcher used to gather results (Sunders, 2012). In data presentation the study made use of tables and pie charts. In data analysis, the study used the SPSS v 26 software. The study used tables and charts because tables make it easy to compare pairs of related values, organize data that have a direct relationship while pie charts display relative proportion of

multiple classes of data and summarize data sets in a visual form. Data was further analyzed using the regression analysis to find the relationship between the variables.

3.11.1 Multiple Regression Assumptions

The study made use of the multiple regression technique in data analysis. A multiple regression analysis is a commonly used techniques due to the simplicity of the application and the ease of which the results are interpreted. The analysis involves a linear curve fitting in its simplest form determining the gradient and constant for a single line equation in the form of $y=mx+x$, where y is the independent variable and x is the dependent variable whilst m represents the proportion of variance in y explained by x (Wegner, 2007). The study made a number of assumption in computing the multiple linear regression (Chiba, 2015, Tomkins, 1991):

- There is a linear relationship between the dependent and independent variables
- The multivariate normality is represented by residual assumed to be normally distributed
- The absence of multicollinearity i.c no highly correlated independent variables. This can be tested using the Variance Inflation Factor (VIF).

3.12 Pilot Study

For the research to incorporate aspects of the validity and reliability of the study, the researcher conducted a pilot test. According to Ismail et al. (2017), a pilot study is a small scale research project that is conducted before the final full scale project is conducted. The pilot study targeted the customer, senior managers and employees not included in the sample but contain the attributes of the final sample. The main aim that the researcher did a pilot test was to check if the data collection tools produced the desired data, establishes the validity of the research based on the expert knowledge of other researchers in the discipline, thus preventing falsified work from being accepted within the area of study and provides valuable feedback that the research revised and improved before the final study.

3.13 Summary

The chapter clearly outlined the methodology that was applied in the research. A positivist research philosophy was adopted. An explanatory research design was employed in the study. The researcher made use of stratified random sampling and targeted Reserve Bank of Zimbabwe, Ministry of Finance and Cryptocurrency firm's employees. The final sample

consisted of 487 respondents. Both questionnaires and interviews were used for data collection in which interviews were scheduled with the RBZ and MoF employees whilst questionnaires were distributed to the cryptocurrency firm's employees. Both primary and secondary data was used to collect data in the study and all procedures were considered so as for the researcher to incorporate reliability, validity and ethical considerations. The next chapter will look at data presentation, discussion analysis.

CHAPTER IV

DATA PRESENTATION, DISCUSSION AND ANALYSIS

4.0 Introduction

The chapter focused on the data presentation, discussion and analysis of the result obtained in the field. The data as presented using tables, pie charts and bar charts and analysed using descriptive and inferential statistics. Descriptive statistics made use of the mean and standard deviation whilst inferential statistics made use of the multiple regression analysis. The data presented sought to answer the research objectives in chapter 1.

4.1 Response Rate

The study made use of stratified random sampling which targeted 298 respondents. The study made use of both the questionnaires and interviews in which the cryptocurrency firms data was collected using interviews whilst questionnaires were used to collect data from the RBZ and Ministry of finance employees. The cryptocurrency were asked questions which were mainly targeted on the conditions that are favourable for them to operate whilst the RBZ and Ministry of finance was asked question mainly in considering the regulation framework that is favourable for cryptocurrency operation in the country. Table 4.1 shows the questionnaire and the interview response rate.

Table 4.1 Questionnaire and Interview Response rate

Administered Questionnaires	Collected Questionnaires	Response Rate
330	300	90%
Scheduled Interviews	Done Interviews	Response Rate
50	30	60%

Source: Primary Data (2021)

Table 4.1 shows the questionnaires and interview response rate for the study. From the 330 administered questionnaires to the MoF and RBZ employees 300 questionnaires were collected successfully error free. From the 50 scheduled interview with the cryptocurrency firms employees and traders 30 interviews were done. According to Roth and Bevier (2018), an overall response rate for an empirical study should be 60% or more so that the results would be valid. The study managed to collect an overall of 90% on questionnaires and 60% on interviews, thus basing on Roth and Bevier (2018) the results obtained from the study are sufficient enough to generalize results of the entire population.

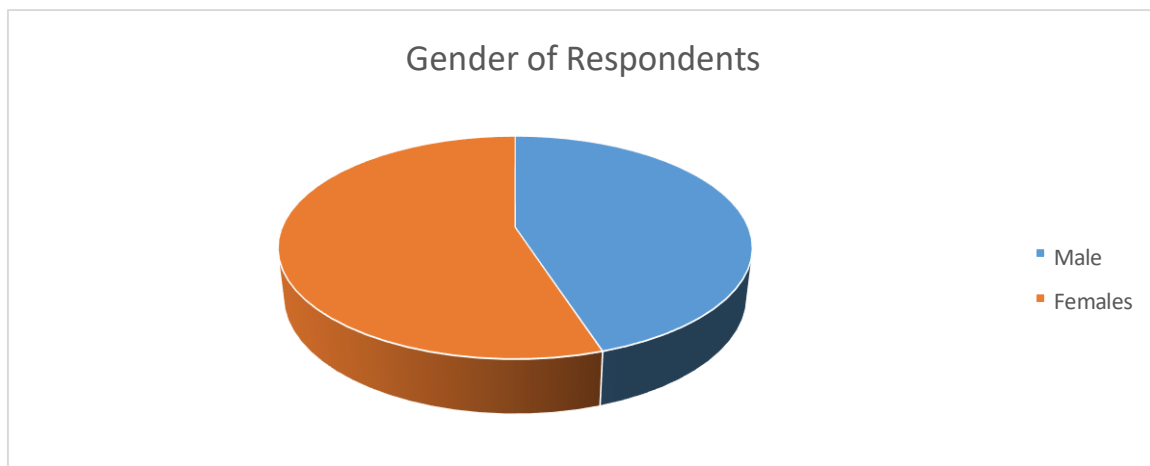
4.2 Study Demographics

This section presents the demographics of the study. The researcher used tables, graphs and pie-charts to show the study demographics.

4.2.1 Gender of the respondents

The respondents were asked to specify their gender on the questionnaires so as to verify if the study has incorporated aspects of gender balance and equity. The incorporation of gender in the research is advantageous in that, the information received would have both views and insights from both genders unlike a gender biased study results which may produce results which are gender biased. Fig 4.1 shows the gender of the respondents incorporated in the study.

Fig 4.1 Gender of the respondents



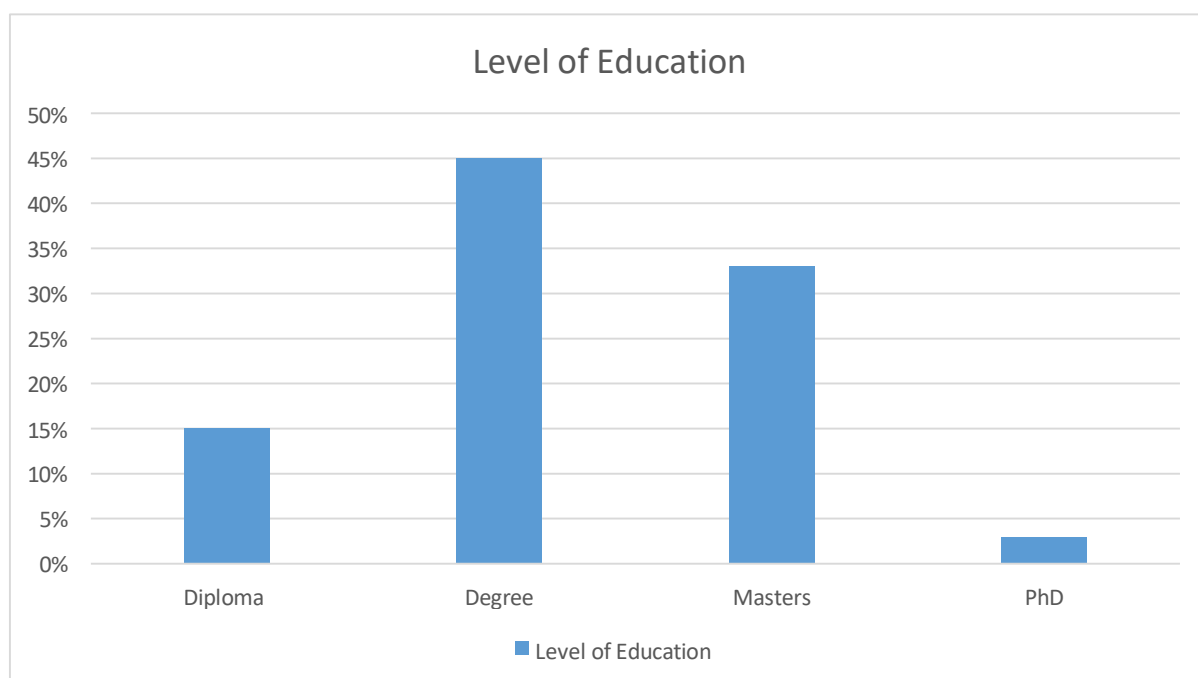
Source: Primary Data (2021)

Fig 4.1 shows the gender of the respondents that were included in the study. From the interviews and questionnaires, 180 respondents were females and 150 respondents were males making a percentile of 55 and 45 respectively. This therefore means that the study included aspects of gender equity which creates a valid and reliable data.

4.2.2 Level of Education

The respondents were asked to specify their level of education, this would make sure that the respondents incorporated in the study understood the questions and are able to answers appropriately. Fig 4.2 shows the level of education of respondents.

Fig 4.2 Gender of the Respondents



Source: Primary Data (2021)

Fig 4.2 shows the level of education of respondents included in the study. The research data, more of the respondents were degree holders consisting of 45% of the respondents. Master's degree holders consisted of 36% of the respondents whilst 15% of the respondents were diploma holders. The PhD holders consisted of the least percentage with only 4% of the study respondents. Overall the study concludes that the respondents include in the study were educated and illiterate which means that they were able to understand the questionnaires and interviews correctly, which enabled the study to collect valid and reliable data.

4.2.3 Number of Years Working in the Organisation

The respondents were asked to specify the number of years they have been working in their respective organisations. The number of years that the respondents have worked in the organisation is useful in the study as the more the number of years that the respondents have worked in the organisation, the more the experience that they have in the organisations operation and the industry at large. This is useful for the study since the respondents might have heard of issues pertaining to the subject area, the previous trend of the investigated information and concerns of the subject matter. This assisted the respondents to collect valid data. Table 4.2 shows the working experience of the respondents.

Table 4.2 Working Experience of Respondents

Reserve Bank of Zimbabwe	Years of Experience	Frequency	Percentages
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	Below 1 year	30	15%
	1-5 years	40	20%
	6-10 years	60	30%
	Over 10 years	70	35%
Total		200	100%
Ministry of Finance	Below 1 year	10	8%
	1-5 years	35	27%
	6-10 years	20	15%
	Over 10 years	70	50%
Total		130	100%
Cryptocurrency Firms	Below 1 year	40	80%
	1-5 years	10	20%
Total		50	100%

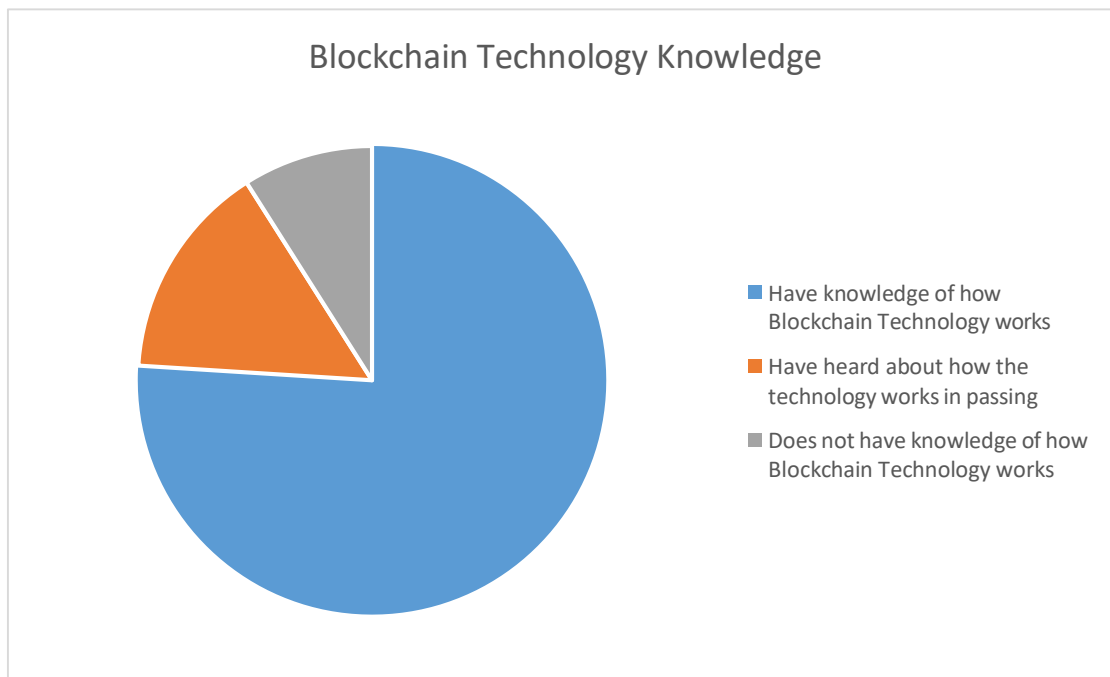
Source: Primary Data (2021)

Table 4.2 shows the working experience of respondents that were incorporated in the study. From the RBZ, most of the respondents included in the study worked for over 10 years which constituted of 35%, 30% of the respondents worked for 6-10 years, and 20% of them worked for 1-5 years whilst 15% of them are below 1 year. From the Ministry of Finance respondents incorporated in the study 50% of them had over 10years experience, 27% of them had 1-5 years or experience whilst 15% and 8% of the respondents had 6-10 years and 1 years and below experience respectively. Respondents from cryptocurrency firms constituted of 20% of respondents in range of experience of 1-5 years whilst 80% of the respondents had below 1 year of experience. This therefore means that the study respondents had the industry experience which gave valid and reliable data for the research.

4.3 Blockchain Knowledge

The respondents were asked if they had knowledge of blockchain technology. This question was asked so as to know if the respondents knew the new innovation since most of the people have not heard of it, basing on the knowledge done form the pilot survey. Knowledge of the function and uses of cryptocurrencies from the respondents is helpful as it gives the study realistic findings based on true facts answered by the respondents. Fig 4.3 shows the Blockchain Technology knowledge.

Fig 4.3 Blockchain Technology Knowledge



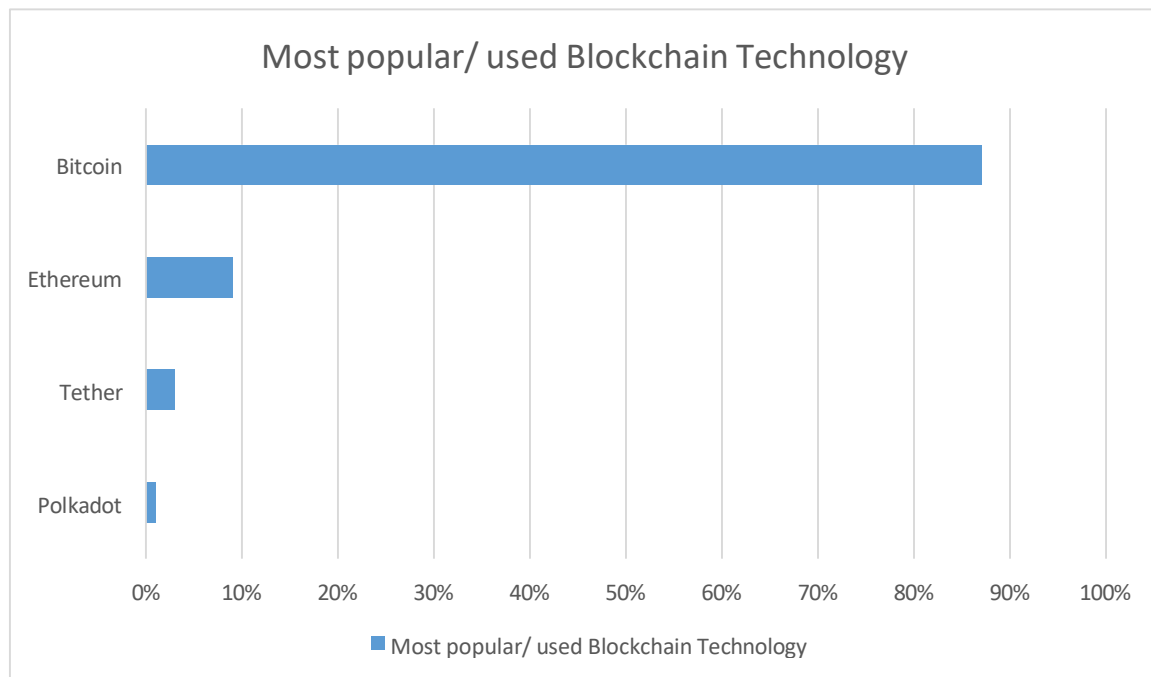
Source: Primary Data (2021)

Fig 4.3 shows the blockchain technology knowledge by the respondents. From the findings, 76 % of the respondents knows how the technology operates, 15% of the have heard of how the technology works in passing and whilst 9% of the respondents do not know how the technology works. From the findings most of the respondents have the knowledge of how the technology works which therefore gives reliable and valid responses to the study.

4.3.1 Type of Blockchain Technology

The respondents were asked the type of blockchain that they have used or which they know. This question what meant to analyse the most popular type of Blockchain technology used in Zimbabwe. Fig 4.4 shows the most commonly or known blockchain technology used.

Fig 4.4 Most popular/ used Blockchain Technology



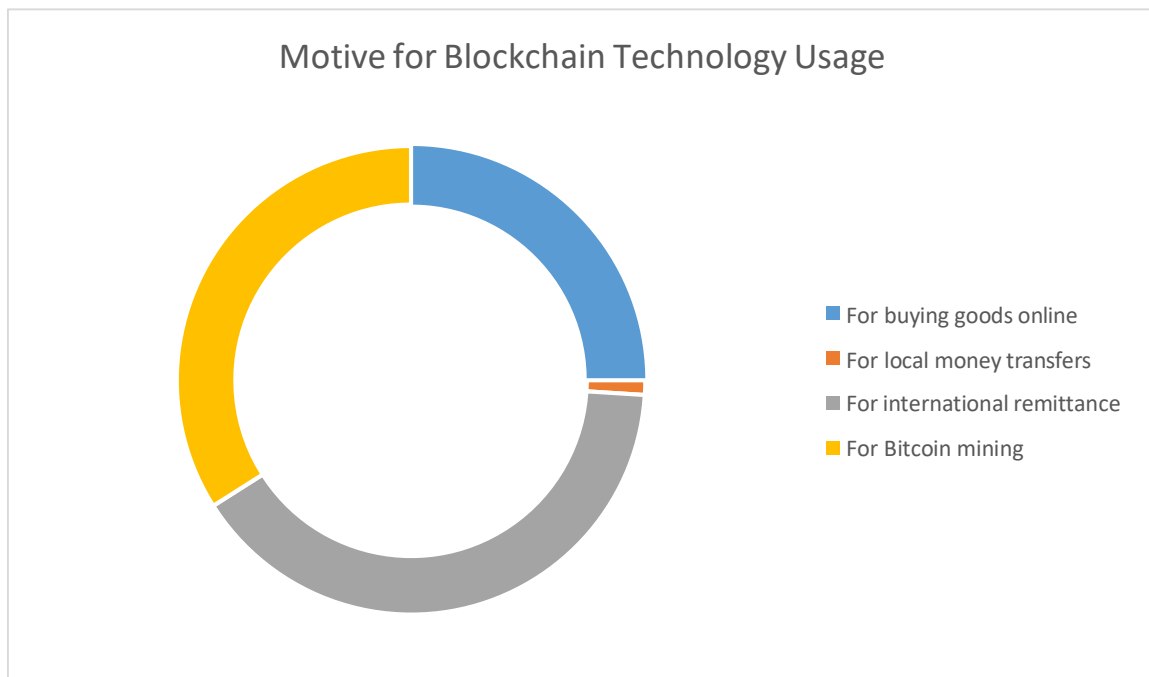
Source: Primary Data (2021)

Fig 4.4 shows the most popular/ used blockchain technology used by the respondents. From the finding most of the respondents know Bitcoin technology with 87%, followed by Ethereum with 9%, Tether with 3% and Polkadot with less than 1%. This shows that most of the respondents only know of Bitcoin which is reflect of its popularity in Zimbabwe and little knowledge of Altcoins is known. This knowledge of coins in use guide the respondents in giving valid and reliable responses.

4.3.2 Motive for using the Blockchain Technology

The respondents were asked the motive if they have used the technology. The respondents were asked this question because it gives the study the knowledge what the technology is being used for in the country. Fig 4.5 shows the motive for using the blockchain technology.

Fig 4.5 Motives for using the Blockchain Technology



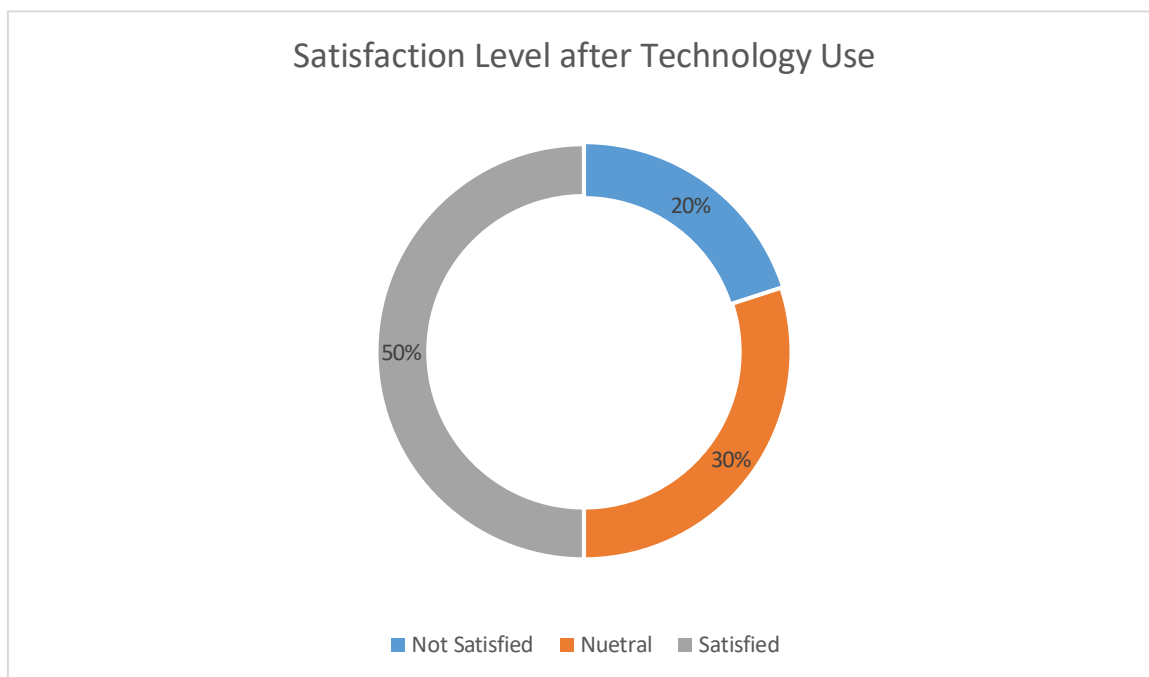
Source: Primary Data (2021)

Fig 4.5 shows the motive of blockchain technology usage. From the findings, most of the blockchain in the country is used for international remittances which constitute 40% of the responses, 34% of the responses said the technology is used for Bitcoin mining, whilst 25% and 1% of the technology is used for buying goods online and for local transfers. This therefore means that most of the people in the country use cryptocurrencies for mining, international remittance and buying goods online. For local money transfers it has not yet been adopted though they are local money transfers such as BitMari in Zimbabwe.

4.3.3 Satisfaction of the Technology

The respondents that have used the technology been asked to specify how satisfied by their use of the technology. The satisfaction rating of the usage of the technology is useful, if most respondents are satisfied that means the technology may be easily be used by others as it proves to be effective in its operation. Fig

Fig 4.6 Satisfaction Level after Technology Use



Source: Primary Data (2021)

From the findings, Fig 4.5 shows the respondents satisfaction levels after the use of the technology. Most of the respondents were satisfied which constituted 50% of the responses, 20% of the respondents were not satisfied whilst 30% of the respondents remained neutral in their responses. This therefore means that the technology may be accepted by the general public if it is allowed to operate in the economy.

4.4 Descriptive Statistics

This section presents data presentation and analysis using descriptive statistics such as the mean and the standard deviation of the responses. The study made use of the SPSS software in the data analysis for each research construct.

4.4.1 Benefits of Cryptocurrency in the Economy

The respondents had been asked to indicate the extent that they agreed or disagreed that cryptocurrency benefits the economy in a certain way. Descriptive statistics relating to benefits that cryptocurrency brings to the economy were computed in Table 4.3 below.

Table 4.3 Benefits of cryptocurrency in the Economy

	N	Mean	Std. Deviation
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R1	Do you believe that cryptocurrency can be used as unit of measure like traditional money?	300	4.7167	0.15747
R2	Cryptocurrencies is highly divisible unlike traditional currencies of up to 8 decimal places. Do you believe that it can solve the divisibility problem that is prone to happen with the use of money?	300	5.0567	0.06955
R3	Cryptocurrency does not have a central authority which means central authority in the network has no power to dictate the rules for owners of Bitcoin. Do you believe than decentralization transaction is more efficient?	300	5.0433	0.17534
R4	Bitcoin also offers a peer to peer cryptocurrency network where there is no master server, which is responsible for all operations, thus the exchange of information (money) is between 2-3 or more software's clients. Do you think anonymity transaction is good?	300	1.4000	.37098
R5	The company has put in place processes that make it easier for the customer to order and buy their products	300	5.1133	.26195
R6	Even if some part of the network goes offline, the payment will continue to operate stable. Do you think cryptocurrency is the solution for system interruptions?	300	4.7733	.39328
R7	Cryptocurrencies usage in the economy stops inflation. The maximum of coins is strictly limited to a certain number, there are neither political forces nor corporations that are able to change that order. Do you believe that cryptocurrency can solve inflation problem in the economy.	300	2.6200	.23730
R8	Cryptocurrencies are easier to register and to use comparing to the account the procedure of opening an accounts in banks. Do you think cryptocurrencies provide financial inclusion to the unbanked members of the society?	300	4.2933	.26590
Valid N (listwise)		300		

Source: SPSS v 26

The respondents were asked certain questions pertaining to the benefits that were to be perceived after the economy has adopted cryptocurrency. From the questions that were asked R1, R6 and R8 shows that the most respondents agreed to the statements, R2, R3 and R5 shows that the respondent strongly agreed to the statements whilst R4 and R7 shows that the respondents disagreed to the statements respectively. These tested results were backed by a lower standard deviation which indicate that the values are clustered close to the mean hence they reliable and valid.

There are many benefits and uses that cryptocurrency brings to the modern economy. These include the currency being used a medium of exchange, stops inflation, the system is not

affected by network interruptions, easy of registration easier to use. From the study findings the respondents believe that cryptocurrencies introduction in the economy can be used as a unit of measure like traditional money ($\bar{x}$ - 4.7167 and σ -0.15747). The respondents agreed to the statement, like traditional money, the cryptocurrencies can be used as a unit of measure like price, thus it can be used in the buying and selling of goods which then become a substitute for traditional money. This result is supported by Barros (2019), who argues that cryptocurrency provides a unit of measure that allows like traditional money the comparison of the value of different goods and services.

Cryptocurrency provides an advantage of divisibility. Unlike the traditional currency divisible to a certain amount, cryptocurrency is divisible up to 8 decimal places. The blockchain technology does not have a central authority thus no one has the power to dictate the rules of the owners. Therefore the respondents strongly agreed to the results of divisibility ($\bar{x}$ - 5.0567 σ -0.0695 and of no central authority ($\bar{x}$ - 5.0433 and σ -0.17534). From the results the respondents appreciate the benefits of cryptocurrency that it brings to the economy that is divisibility in transactions, thus the user can buy anything from cents and this is another characteristics of good money. The unavailability of no central authority in cryptocurrency has a benefit in that the value of the currency may not be affected by government fiscal policies and political pressure which may result in a stable currency.

Blockchain technology offers a peer to peer network where there is no master server which is responsible for all operation thus the exchange of money is between 2 or more software clients which issues anonymity of transaction (Li and Wang 2017). Even if the network goes offline, the system would not be affected by network interruptions. The respondents strongly disagreed to the statement that anonymity of transaction is beneficial in the economy ($\bar{x}$ - 1.4000 and σ -0.37098), from the interviews they reflect that this may cause illegal transaction in the financial space giving an example of Silk Road scandal were an infamous dark web black market sold practically everything including fake passports, illegal weapons, hacking software and dangerous chemical in 2013. The respondents agreed to system stability as a benefit to the economy ($\bar{x}$ -4.773 and σ -0.39328). Modern technology such as mobile money transfer, real time gross domestic settlements and telegraphic transfers are mostly affected by network disruptions thus cryptocurrency usage might solve the problem of network instabilities in an economy (Barros, 2019).

The respondents were asked if they believed that cryptocurrency usage in an economy stops inflation because the maximum coins are strictly limited to certain number, there are neither political forces nor corporation that are able to change that order. The respondents agreed to the statement ($\bar{x}$ -4.2933 and σ -0.2659). This therefore means that in the Zimbabwean economy where inflation is prevalent, cryptocurrencies might be able to stabilize the economy by solving the inflation crisis (Kampl, 2014). Companies have to put in place processes that makes it easier for customer to order and buy their products using cryptocurrency. The respondents strongly agreed to the statement ($\bar{x}$ -5.113 and σ -0.26195). This therefore means that online stores like Ownai.com, Tengai.com and have to put in place the Bitcoin payment method as a way of payment so as to speed up the acceptance of the technology.

Cryptocurrency is believed to financially include the excluded part of the society in financial service sector. Most of the people in the rural areas are financially excluded, 5 out of 15 do not have a bank account. This is due to the fact that a bank needs account maintenance fee, in Zimbabwean instance these are assured by a monthly salary which the rural populace does not have and citizen registration status which some people does not have. Cryptocurrency is easier to register as compared to opening a bank account, it works in minutes and doesn't not require any identity particulars. This might financially include the unbanked members of the society which due to the no availability of the barriers (Fahmy, 2018). The respondents agreed to the statement ($\bar{x}$ -4.2933 and σ -0.26590).

4.4.2 Creating a Regulatory Framework

The respondents had been asked to indicate the extend that they agreed or disagreed that they is need for a regulatory framework that guide the usage of cryptocurrency in the economy. Descriptive statistics relating to question related to creating a regulatory framework were computed in Table 4.5 below

Table 4.5 Creating a regulatory framework

		N	Mean	Std. Deviation
D1	Should the government use cryptocurrencies as a legal currency	300	1.5233	.31593
D2	Should the government use cryptocurrencies a virtual currency	300	5.2534	.36640
D3	Should the government amend its tax laws so as to include cryptocurrency in the taxation system for example included in the Income Tax?	300	5.3356	.26523
D4	Can the government allow the selling and buying of goods using cryptocurrency although not allowing it to be a legal tender	300	5.1033	.23929
D5	The RBZ has to consider cryptocurrency trading and review its Crime and Terrorism Act to treat virtual currency as a money service business for the purpose of anti-money laundering law	300	4.5333	.17445
D6	The government has to review on securities laws so as to include Initial Coin offering and blockchain fund by publishing guidelines on its regulatory treatment in order to lure foreign direct investments	300	5.0000	.10200
D7	In order for the public to gain acceptance of the cryptocurrency the government have to make payment for their services with the option of Bitcoins	300	4.7767	.31718
Valid N (listwise)		300		

Source: SPPSv23

The respondents were asked certain questions pertaining the desirable regulatory framework for cryptocurrency operations in the country. From the questionnaires that were asked D3, D4 and D6 shows that the respondents strongly agreed to the statements whilst D5 and D7 agreed to the statements. D1 responses shows that the respondents strongly disagreed with the statement. These tested results were backed by a lower standard deviation which indicate that the values are clustered close to the mean hence they reliable and valid.

Most of the government globally have legalised cryptocurrency as a virtual currency rather than a legal tender. Most of the governments argue that cryptocurrency is decentralised and does not need a central authority to regulate it, thus the government has its sore responsibility of issuing a legal currency and controlling the money supply. This therefore means that cryptocurrency cannot be a legal tender since the government doesn't does not have the power to control it and to issue it thus the qualities of good money. The respondents disagreed that cryptocurrency be used as a legal currency ($\bar{x}$ -1.5233 and σ -0.31593) rather be used as a virtual currency of which the respondents strongly agreed to it ($\bar{x}$ -5.2534 and σ -0.36640). The cryptocurrency cannot be used as a legal currency since are non-controllable and not mandatory

means of cancelling debts rather can be used as a thing or good which can be used as a means of barter trade between a transaction like how the Argentinian government views it (Lauonia, 2018).

The government have to allow the buying and selling of goods using cryptocurrency although not considered as a legal tender. This then enables it to be liable for taxation in the country. The respondents strongly agreed that the government should let cryptocurrency be used as a currency for buying and selling ($\bar{x}$ -5.1033 and σ -0.23323). The respondents also strongly agreed that the government have to include transaction involving cryptocurrency in the tax bracket such as income tax ($\bar{x}$ -5.3356 and σ -0.26523). A law have to be put in place so that the tax the transaction that use cryptocurrency before the cryptocurrency is allowed to be used as a currency for buying and selling. This also can give revenue to the government as an advantage. This can be mirrored to Switzerland were the government has levelled cryptocurrency for income and wealth tax.

Most of the reserve banks globally are reviewing their financial crime and terrorism act so as to treat virtual currencies as a money service business so that it cannot go against the anti-money laundering laws. Countries like Canada have amended its Proceeds of Crime and Terrorism Act to treat virtual currencies as a money service business (Ahmad, 2014). In Mexico they have enacted a law extending the application of its laws regarding money laundering to virtual assets by reporting transactions exceeding certain amount to the central bank. The respondents agreed that the government have to issue laws that that protect cryptocurrency operations and not be recognised as financial crime and terrorism.

Worldwide small businesses are being financed by initial coin offering (ICOs), as the new normal an ICO is a type of funding using cryptocurrencies which is often in a form of crowdfunding were a quantity of cryptocurrency is sold in the form of 'tokens' (coins) to speculative investors in exchange for a legal tender or other (generally established and more stable) cryptocurrencies such as Bitcoin or Ethereum. Therefore the governments have to review on their securities law so as to include ICOs and Blockchain funding by publishing guidelines on its regulatory framework in order to lure this types of investments. The respondents strongly agreed to the statement ($\bar{x}$ -5.000 and σ -0.10200). This therefore means that the government has to put in place laws that govern ICOs. The Germany BaFin has already published information on the regulatory assessment of ICOs and token and its Federal Financial Supervisory Authority has stated that firms involved in ICOs need to assess on a case-by-case

basis whether the ICOs qualify as financial instruments (transferable securities, units in collective investment undertakings or investments) or as securities and therefore trigger the need to comply with relevant legislation.

In order to normalize the new normal of buying and selling using cryptocurrencies, the government have to put on its payment platforms cryptocurrency as a means of payment of their services. The government for example may put the cryptocurrency as an alternative in the payment of passport application, municipal bills or payment of state university fees. This may push the public to be aware of the blockchain technology and to be interested in using them. Countries like Switzerland are allowing the payment of municipal bills using cryptocurrencies up to a maximum amount of \$210 (Kunton Zug, 2017).

4.4.3 Cryptocurrency, liquidity and cost of international remittances

4.4.3.1 Liquidity Concerns

The respondents has been asked to indicate the extend that they agreed or disagreed that cryptocurrencies may solve the liquidity cruch that keeps on emanating in the economy. Descriptive statistics relating to cryptocurrency and the liquidity crunch is computed below in Table 4.5

Table 4.5 Cryptocurrency and Liquidity Crunch

		N	Mean	Std. Deviation
A1	If cryptocurrency is recognised as a virtual currency can be able to solve the liquidity crunch cryptocurrency as a way of payment and settlement of debts thus people would not demand more of the hard currency	300	4.6771	0.12747
A2	Cryptocurrency usage in the economy have the ability to stop inflation in the economy	300	5.3640	0.14420
A3	The government RBZ have to encourage the banks to include cryptocurrency services as well as the introduction of Bitcoin ATMs	300	5.0433	0.27534
A4	Cryptocurrency cannot be traded like other currencies at an exchange rate. Thus cryptocurrency can solve the money changing problem in the economy	300	3.4000	.170980
A5	Most of the rural populace in Zimbabwe. Cryptocurrencies provide users easier registration and lower operating costs unlike physical cash there is no need to pay commission and fees to banks.	300	5.1133	.36195
Valid N (listwise)		300		

Source: SPSS v26

The respondents were asked question related to cryptocurrency and liquidity crunch. In A1 and A2 the respondents agrees with the statement whilst in A3 and A5 the respondents strongly agreed to the statement asked. Respondents on A4 remained neutral in their responses. All the results had a lower standard deviation which shows that the results are valid and correct. A lower standard deviation means that the responses are clustered close to the mean.

Most of the governments globally has recognized cryptocurrencies as a virtual currency rather than a legal tender (Canada, Switzerland and Germany), however they have allowed it to be used as a medium of exchange between goods and services. Most of the demand of a fiat currency by consumers is mainly due to buying of commodities. If the government allows the cryptocurrency to be used as means of buying goods and services, the public may have an option of buying either with cash of cryptocurrencies. This eases the cash demand that usually occurs in the economy, they by solving the liquidity crunch. Most of the respondents agreed to the statement ($\bar{x}$ -4.6771 and σ -0.12747). This therefore means that most of the respondents appreciate cryptocurrency to be a virtual currency that can be able to solve the liquidity crunch in the economy (WorldPress, 2012).

Most of the Central Banks globally are now appreciating cryptocurrency in their economies and therefore are now encouraging banks to look at the technology and to include this technology in their service offerings. In Britain for example they have allowed Bitcoin ATM to be included in the financial system and these Bitcoin ATMs have sprouted around Europe (Times, 2018). This has also seen hundreds of merchant's accepting cryptocurrency payments indifferent economies. Therefore an economy like Zimbabwe which have liquidity concerns may have to consider cryptocurrency adoption. The respondents strongly agreed ($\bar{x}$ -5.0433 and σ -0.27534) that the RBZ have to encourage banks to offer cryptocurrency related services since it have become the new normal in the financial industry.

Cryptocurrency like other fiat currencies that are traded with an exchange rate, this technology cannot be traded like other currencies at a current exchange rate. In Zimbabwe, the country's currency is being inflated and depreciated constantly due to the black market that can raise or lowers the exchange rate at whatever rate they want depending on the forces of demand and supply. This causes the inflationary pressures to the country's currency. Am economy like Zimbabwe have to introduce cryptocurrency since it is no traded with other currencies, thus is less likely to work in the black market system. Most of respondents were neutral in their

responses ($\bar{x}$ -3.400 and σ -0.17098). Therefore this calls for an investigation to if the cryptocurrency and solve the black market crisis.

According to Chikwindo *et al.*(2014), the levels of financial inclusion in the country are lower in the rural areas despite the fact that 65% of the rural populace lives in the rural areas. From his research, they found out that most of the rural populace did not have a bank account because banks require a monthly servicing fee for the account thus needs the assurance of a salary for which most of the populace do not have. Cryptocurrencies provides easier user way of registration, easier usage and lower operating cost as compared to a bank account which require monthly fees and payment of transaction costs. Therefore if cryptocurrencies is introduced in the economy is believed to financial include those excluded in the financial service sector. The respondents strongly agreed to the statement ($\bar{x}$ -5.1133 and σ -0.36195), thus are in support of cryptocurrency adoption.

According to Jonker (2017), the cryptocurrency technology usage in an economy have the ability to lessen the effect of inflation. This is due to the fact that the maximum number of coins is strictly limited to a certain number, there are neither political forces nor corporations that are able to change that order, and thus there is no possibility for the development of inflation in the system. The respondents strongly agreed ($\bar{x}$ -5.3640 and σ -0.14420) that inflation has the ability of solving the inflationary pressures that are happening in the economy.

4.4.3.2 Cost of Remittance

The respondents were asked to indicate the extend that they agreed or disagreed that cryptocurrencies may lower the cost of remittances in the economy. Desriptive statistics relating to the cost of remittance is shown in Table 4.6 below

Table 4.6 Cost of Remittance

	N	N	Std. Deviation
Q1	The cost of remittance is high using financial services as compared to cryptocurrencies. Should the government allow cryptocurrencies remittances since there cost less	300	.21593
Q2	Reducing the cost of remittances would results in more money remaining in the pockets of the migrant families and would have a significant effect on the income levels of remittance families	300	.30929
Q3	An increase in consumption, savings and investments in the local economies would lead to economic growth	300	.07445
Q4	Many migrants have faced difficulties to obtain the necessary identification documentation to enter the financial mainstream thus facing difficulties in sending remittances. Cryptocurrency does not require identification and transaction are faster, cheaper and efficient	300	.33562
Valid N (listwise)		300	

Source: SPSS v26

The respondents were asked question related to cryptocurrency and the cost of remittances. In all the question, the repondents strongly agreed to them. The responses were backed with a lower standard deviation which means that the response are clusterd close to the mean thus thy are valid and reliable.

The Zimbabwean economy has been backed with remittances over the past years due to a lot of migration of people to developed countries. The value of personal remittances received in Zimbabwe have to around \$2 billion whilst the cost of remittance has reached an estimate of around \$49 million (World Bank, 2018). Therefore this signals that the cost of remittances is high using financial services which at times takes 15% of the total amount of money sent. As compared to send money remittance services, on average transaction fee for cryptocurrency is 4.355% for an international transaction. This therefore makes an average of 12% different between the two services. The respondents strongly agreed ($\bar{x}$ -5.5233 and σ -0.21593), that the government should allow cryptocurrencies remittance services to operate in the country as it reduces the remittance cost.

If the sender of the remittance is charged a low cost of remittance fee, this therefore results in more money remaining in the pockets of the migrant families which would have an effect on the income levels of the receiver. An increase in the income level for the receiver of the remittance would shift the demand curve to the right as the receiver increase his/her demand

of goods and services. All of this may result in economic growth as the income levels of the people increase. The respondents strongly agreed ($\bar{x}$ -5.1033 and σ -0.30929), that the lower cost of remittance fee, the more the income received. The respondents further strongly agreed ($\bar{x}$ -5.533 and σ -0.07445) that if the income increases, consumption, savings and investments in the economies would lead to economic growth.

Many migrants have faced difficulties to obtain the necessary identification documentation to enter the financial mainstream thus facing difficulties in sending remittances. Cryptocurrency has become the answer to this problem, it offers a new remittance solution that does not necessarily require user identification. The technology uses a combination of public and private keys on the open-source software (Jonhson, 2018). Such development would allow individuals to send money without any identification needed for example a bank account which needs a permit for foreigners to open an account. The respondents strongly agreed ($\bar{x}$ -5.2250 and σ -0.33586), that lack of identity documents have made other migrants not to send remittances thus cryptocurrency has solved the problem.

According to interview data, they argued that the technology has unlimited amounts of transaction's that is, each of the wallet holders can pay to anyone, anywhere and any amount. The transaction are not controlled or prevented thus a user can make transfers anywhere in the world wherever another Bitcoin wallet is located. That it offer no transaction boundaries. The payments made in the system are impossible to cancel, the coins cannot be faked, copied or spent twice thus these capabilities guarantee the integrity of the entire system and allowing online shops, resources and companies to accept Bitcoin globally (Jonker, 2017).

4.4.4 Opportunities and threats of cryptocurrencies in the economy

The respondents were asked to indicate the extent they agreed or disagreed that cryptocurrencies has its own opportunities and threats that the economy enjoy/ injure from the adoption of cryptocurrency. Table 4.7 shows the opportunities and threats that follows after the economy has adopted cryptocurrencies.

Table 4.7 Opportunities and Threats of cryptocurrency adoption in an economy

		N	Mean	Std. Deviation
W1	Bitcoin can help the unbanked part of the community to have peer to peer transactions which was not possible with traditional banking technology	300	5.7167	0.25747
W2	Bitcoins technology allows for business transaction to occur quickly and are easier to use	300	4.0567	0.16955
W3	The government can make revenue through taxation if cryptocurrencies are taxed for example Income Tax, Value added tax.	300	5.0453	0.18534
W4	The government can treat cryptocurrency has a commodity like Gold. Cryptocurrencies could potentially become a big player in the commodity market.	300	3.4223	0.33098
W5	Bitcoin has strength by design to make it a viable currency that has elevated it in status over the years, more notably the fixed limit of Bitcoin that will exist, Bitcoin will be mined with diminishing returns	300	5.1233	0.42195
W6	The value fluctuations that plague cryptocurrencies puts doubt in users, as well as investors	300	4.7733	.17328
W7	Cryptocurrencies have also seen fraud and theft, generally due to faulty system setups by exchange companies.	300	5.6200	.23730
Valid N (listwise)		300		

Source: SPSS v26

The respondents were asked the extend they agreed or disagreed to the statements related to the threats and opportunities that are derived from the adoption of cryptocurrency. The respondents strongly agreed in statements W1, W3 and W7 whilst they agreed with the statement on W2 and W6. Some of the respondents remained neutral in their responses W4. All the responses have a lower standard deviation which means that the results showed clustered close to the mean thus thy are valid and reliable.

According to Kelly (2014), transformative technologies start by solving ga specific problem in the industry, for example cryptocurrency has been poised to help remediate the problems related to unbanked consumers. Significant portions of the population in developing countries are unbanked. In Latin America, 60% of 600 million inhabitants have no access to bank accounts (Magro, 2016). Bitcoins technology allows for individuals to exchange currency without needing a third trusted party, like a bank, to oversee the transaction. All that is needed to use Bitcoin is a mobile phone, which 70% of Latin Americans do have access to (Magro, 2016).

The respondents strongly agreed ($\bar{x}$ -5.7167 and σ -0.25747) that the ad-hoc networking capability, two users can trade Bitcoin with each other by scanning QR codes displayed on their phones printed out by the application truly solution has solved a problem that has existed for many years for some people.

Bitcoin technologies have allowed transactions to be faster than before, both international and local transfers may be wired within minutes to the receiver which had made individuals and business transactions to occur quickly. The respondents agreed to this benefit ($\bar{x}$ -4.0567 and σ -0.1695) as the technology sends transaction within minutes. The transaction sent will be having a lower transaction fee than bank transfers. The peer-to-peer system allows all the all the information about the transaction in the blockchain network shared (like how, when), but there is no data about the receipt nor the sender of the coins (there is no access to the personal information of the owners wallet. Thus blockchain transaction are fully anonymous and at the same time fully transparent (Li and Wang, 2017).

Globally governments are making fiscal by taxing the cryptocurrency, like Canada which is changing Income Tax on all transaction done by cryptocurrency and German which is taxing the ICOs, governments may be able to increase its revenue. The respondents strongly agreed ($\bar{x}$ -5.0453 and σ -0.1854), taxing cryptocurrency operation increases the government revenue. Tax laws regarding the taxation of cryptocurrencies are required before digital currency could be considered a truly valid form of transactions in an economy. Like the European Court of Justice announced that it viewed Bitcoin transactions as exempt from value-added tax (Hileman, 2016).

Most of the governments are recognizing the important of cryptocurrency as commodity like Gold, thus the technology could became a big player in the commodity market. Similar to which have a volatile value that increase and decrease due to economic effects, the cryptocurrency has seemingly mimic the characteristics of gold. Gold has been a long standing holder of value and is based on the universal acceptance and trust of its value, cryptocurrencies could potentially become a big player in the commodity market. Cryptocurrencies have a unique attribute of being purchased through direct online mechanism, which creates entry for buyers thus if Bitcoin continues to be a refugee for inflating currencies, it would gain validity to investors and push deeper into becoming in the financial mainstream. Most of the respondents were neutral in their responses ($\bar{x}$ -3.4223 and σ -0.33098), which means that they were not sure if the system can work our developing economy.

According to King (2013), Bitcoin has strength by design to make it a viable currency that has elevated it in status over the years, more notably the fixed limit of Bitcoin that will exist thus Bitcoin will be mined with diminishing returns every four years until the maximum number of Bitcoins are reached. This aspect of Bitcoin is important for its value. The respondents agreed to the statement ($\bar{x}$ -5.1233 and σ -0.42195), they believe that due to the limited amount of Bitcoin, it would never be inflated and generally cryptocurrencies are generally regarded as being protected from inflation originating from national government changes or restrictions thus create a safe haven for investors to put their wealth into, as it does not lose value based on inflation.

However, cryptocurrencies have demerits in its designs such as the issue of value fluctuation and high risk of fraud and theft. The value fluctuations that plague cryptocurrencies puts doubt in users, as well as investors. Ultimately a limiting factor in cryptocurrency is general acceptance. The respondents agreed to the statement ($\bar{x}$ -4.7733 and σ -0.17328). Value fluctuations reduce trust that a consumer's value would be retained on a day to day basis, limiting faith in the currencies overall worth. The lack of central ownership of cryptocurrencies means that any attempt to remediate this marketing problem using advertisements could theoretically help the investing company's competition. This is not an ideal situation for a marketing plan (PwC, 2019).

The lack of central ownership of cryptocurrencies means that any attempt to remediate this marketing problem using advertisements could theoretically help the investing company's competition. This is not an ideal situation for a marketing plan. The respondents strongly agreed to the ($\bar{x}$ -5.6200 and σ -0.23730), thus as long as cryptocurrencies remain in an area not generally covered by law, user acceptance will be limited. User's need to trust that any transactions using cryptocurrencies are legal and binding. The markets and governments are being slow to react to the new technology. Ultimately, all of these factors limit consumer's trust in Bitcoin and cryptocurrency (Bearman, 2015).

4.5 Inferential statistics

This section presents the analysis of data using model the multiple regression analysis.

4.5.2 Multiple Regression Analysis

4.5.3 Validity and Reliability of the Model

4.5.3.1 Multiple Regression Assumptions

In order to perform a valid regression analysis, certain assumptions must be met. In addition, a sample size of at least 20 cases per dependant variable is required (Wegne, 2007), thus our sample size exceeds the minimum. The assumption of normality and present of outliers, a linear relationship and multicollinearity needs to be verified.

Normality is required to ensure that biases are not introduced in the data analysis. Using SPSS Skewness and Kurtosis scores are calculated in Table 4.8 below. According to Field (2013), magnitudes of each static should not exceed 2. From the table, all values are within the recommended range.

Table 4.8 Skewness and Kurtosis statistics

		R1	R2	R3	R4
N	Valid	300	300	300	300
	Missing	0	0	0	0
Skewness		-1.083	-0.719	-0.168	-0.589
Std. Error of Skewness		.041	.041	.041	.041
Kurtosis		1.128	0.451	-0.341	0.245
Std. Error of Kurtosis		.081	.081	.081	.081

Source: SPSS v26

In the data computed in Table 4.8 study variables named R1, R2, R3 and R4 represent the study variables that is government revenue, economic variables, liquidity crisis and cost of remittances respectively .The data was visually inspected for outliers using the SPSS boxplots. Some outliers were identified, however since the mean does not appreciably differ from the trimmed mean, these are adjudged to not affect the data (Pallant, 2013). These therefore means that the data set is well modelled by a normal distribution and this also shows that the random variable underlying the data set is normally distributed.

The researcher also used the Variance Inflation Factor (VIF) and normality test detect multicollinerity in the statistics. The methods used to check validity will be summarized in Table 4.9.

Table 4.9 Validity data for the sample

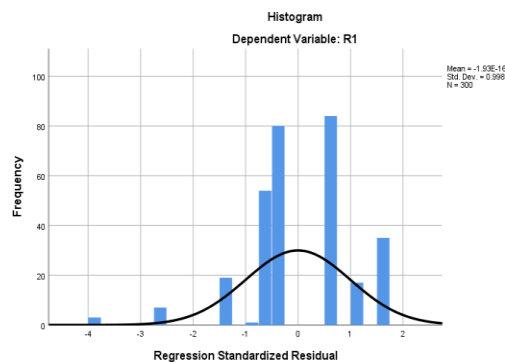
Table 4.9.1 Multicollinearity statistics

		Collinearity Statistics	
Model		Tolerance	VIF
1	R1	0.351	2.846
2	R2	0.512	1.954
3	R3	0.773	1.293
4	R4	0.367	2.725

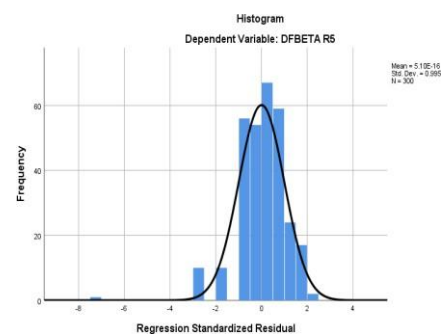
4.9.2 Normality

Fig 4.7 Normality Tests

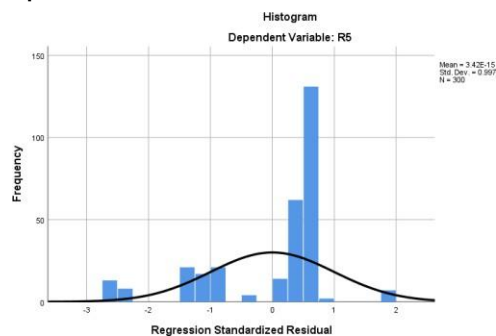
Dependent Variable: Government Revenue



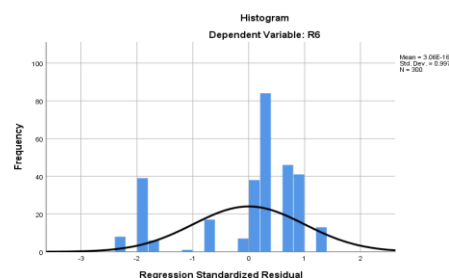
Dependent Variable: Liquidity Crunch



Dependent Variable: Cost of Remittances



Dependent: Economics Variables



To detect multicollinearity in the study variables the researcher also made use of the VIF and the normality test. The VIF quantifies the severity of multicollinearity in an ordinary least squares regression analysis that it provides an index that measure how much the variance of an estimated regression coefficient is increased because of collinearity. Values that exceed 10 are often indicating multicollinearity however the study obtained 2.5846 on government revenue, 1.954 on liquidity crunch, 1.293 on cost of remittances and 2.725 on economic variables. Therefore the model best fits and there is no multicollinearity in our variables. A

normal distribution of residuals around the predicted scores demonstrates that the assumption of multivariate normality holds.

4.5.3.2 Reliability

Assessing reliability involves checking for internal consistency of the model constructs. The study used Cronbach's Alpha to check the reliability of the statistics. According to Field (2013), the Cronbach's Alpha should be above 0.7. Scores that are below this level have to be excluded from further analysis. Table 4.10 shows the reliability statistics for model constructs.

Table 4.10 Reliability statistics for model constructs

Constructs	Cronbach's Alpha Based on Standardized Items ^a
Government Revenue	0.747
Liquidity Crunch	0.810
Cost of Remittances	0.891
Economics Variables	0.738

The study variables range from 0.738 to 0.891 which indicate internal consistency of the measurement instrument.

4.4 Results per Hypothesis

The study seems to find predictor variables of the adoption of cryptocurrency (AoC) in the economy and its effect on the modern economy. The constructs that the study is using are government revenue, liquidity crunch, cost of remittances and economic variables. The respondents were asked questions pertaining the study constructs, descriptive statistics for these constructs are provided in Table 4.11 below.

Table 4.11 Descriptive for study

	Mean	Std. Deviation	N
Government Revenue	4.5233	.91593	300
Liquidity Crunch	4.1033	.93929	300
Cost of Remittances	4.5333	.57445	300
Economics Variables	4.7767	.41718	300

Source: SPSS v 26

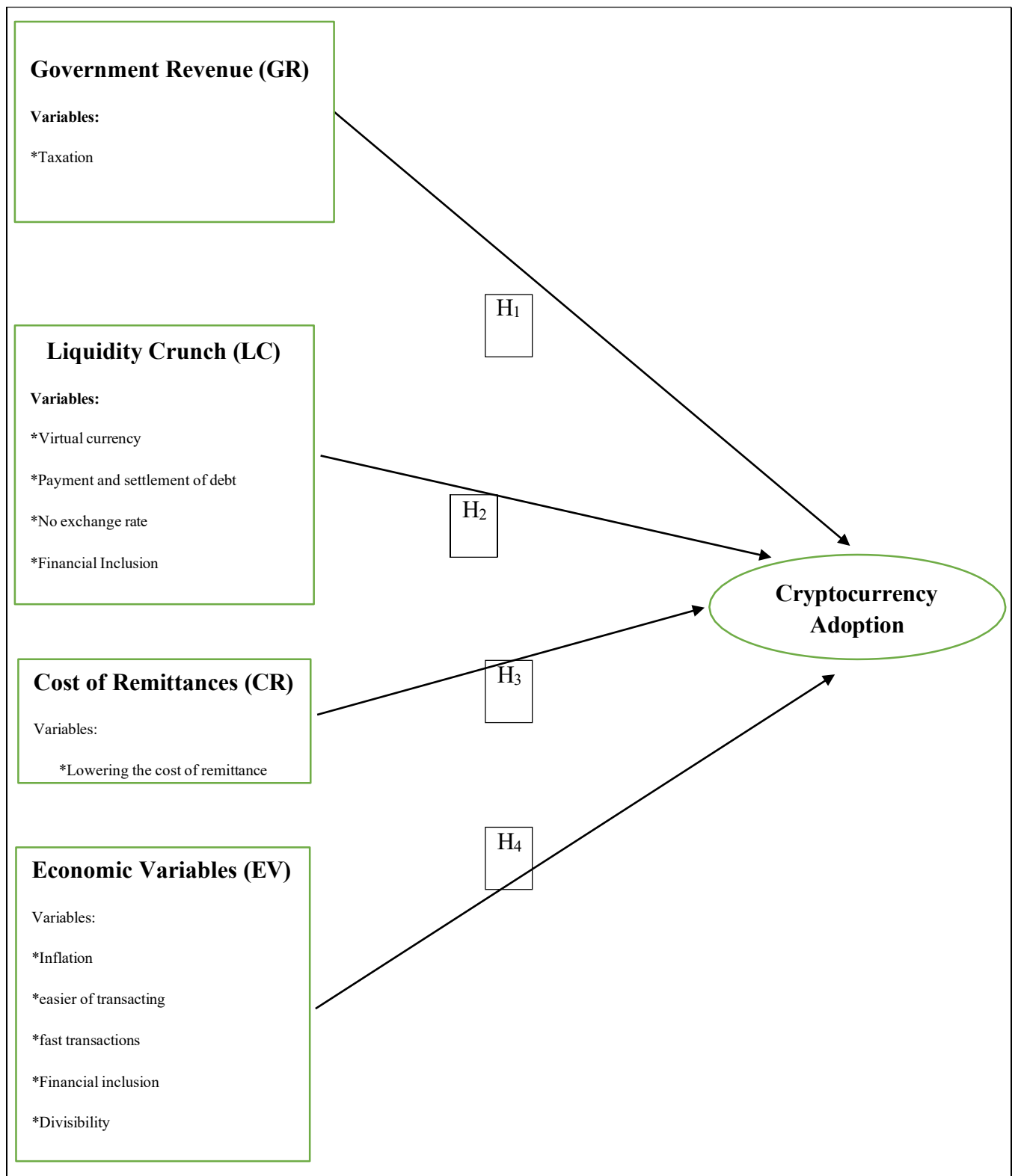
The table indicates that most of the respondents agreed with the statement, this therefore means that the respondents agreed to the statement that the adoption of cryptocurrency (AoC) have an

effect in increasing government revenue, solving the liquidity crisis, lowering the cost of remittance and increasing the economic variables.

4.4.1 Cryptocurrency adoption

The study independent variable that is the cryptocurrency adoption in the economy has brought about hypotheses that are being investigated in the study. This therefore forms a model in Figure 4.8 below which summaries the subset of the hypothesised model being analysed.

Fig 4.8 Independent predictor variables of cryptocurrency adoption in the hypothesised model.



The researcher analysed the correlation between study constructs. The Bivariate Pearson Correlation analysed is shown in Table 4.15 below. A bivariate correlation value of -1 shows

a positively negative linear relationship, 0 shows no relationship and +1 shows a perfectly positive linear relationship.

Table 4.15 Correlation matrix for cryptocurrency adoption as a dependent variable

	R1	GR	LR	EV	CR
R1	0.663**				
GR	0.614**	0.168*			
LC	0.440**	0.629**	-0.314**		
EV	0.700**	0.624**	0.383**	0.754**	
CR	0.664**	0.562**	0.433**	0.713**	-0.709**

**Correlation is significant at the 0.01 level (2-tailed).

*Correlation is significant at the 0.05 level (2-tailed)

The bivariate Person correlation shown in Table 4.15 shows the correlation matrix for the dependent variable which was correlated with the independent variable of the study. From the findings (GR) government revenue (0.663) and (EV) economic variables has shown as a positive relationship to cryptocurrency adoption whilst (LC) liquidity crunch and (CR) cost of remittances has shown a negative relationship to cryptocurrency adoption

The study also employed a linear multiple regression model which shows that the four predictor variables. Table 4.16 shows summary statistics of regression model with cryptocurrency adoption as the dependent variable.

Table 4.16 Summary of statistics of regression model with adoption of cryptocurrency (AoC) as the dependent variable

Model Summary ^a				
Model	R	R Square	Adjusted R Square	Std. Err of the Estimate
1	.785 ^a	.616	.605	.53337

Predictors: GR, LC, CR, EV

Table 4.16 shows the correlation coefficient (R), the coefficient of determination (R^2) and the adjusted R^2 . The four predictor variables were used. The results in R shows that there is a perfectly linear positive relationship between adoption of cryptocurrency (AoC) and economic

growth. This is also supported by a R squared of 0.616 and an adjusted R² of 0.65 which shows that the data is closer to the mean.

Next, a test for significance was necessary using the ANOVA technique so as to show the goodness of fit of the model. Table 4.17 shows the Anova results for regression model using adoption of cryptocurrency (AoC) as the dependent variable.

Table 4.17 ANOVA results for regression model adoption of cryptocurrency (AoC) as the dependent variable

		ANOVA Table				
		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	82.129	4	16.426	57.740	.000 ^b
	Residuals	51.206	295	.284		
	Total	133.335	299			

a. Dependent Variable: ADC

b. Predictors: GR, LC, CR, EV

From table the ANOVA table, we test the significance of the test results (Chiba, 2015). The ANOVA table above represent the results above that the model is a good fit and is statistically significant with p-value less than 0.001 (F= 57.740).

The effect of each predictor variable can be analysed using the table of coefficients below in Table 4.18.

Table 4.18 Regression coefficient results and hypothesis decisions with adoption of cryptocurrency (AoC) as a dependent variable

		Coefficients					Hypothesis Decision
		Unstandardized Coefficients		Standardized Coefficients			
Model		B	Std. Error	Beta	t	Sig.	
1	(Constant)	0.476	0.208		2.291	0.023	
	GR	0.121	0.088	0.107	2.379	0.170	H1 Accepted
	LC	-0.115	0.048	0.168	1.207	0.002	H2 Rejected
	CR	-0.334	0.077	0.332	1.349	0.000	H3 Rejected
	EV	0.291	0.095	0.243	3.063	.0003	H4 Accepted

a. Dependent Variable : AoC

The regression equation is therefore:

$$AoC = 0.476 + 0.21(GR) + -0.115(LC) + -0.334(CR) + 0.291(EV) + 0.208$$

Summary of the results

The study analysed the effect of the adoption of cryptocurrency and government revenue. The study made use of tax as a variable for government expenditure. From the analysis there was a positive relationship which means if the economy adopts cryptocurrency, the government can be able to gain revenue through taxes that are taxed on the buying of goods using cryptocurrency, ICOs and international remittances. This has an effect on government revenue increases due to the taxes charged by the government which has resulted in ($\beta = 0.121$, p-value < 0.01) therefore the study conclude that:

H1- There is a positive relationship between cryptocurrency adoption and an increase in government revenue

The study analyses the effect of the adoption of cryptocurrency and the liquidity crunch in the economy. The adoption of cryptocurrencies has lessened the effect of liquidity crunch in the economy through it variables which are the introduction of a virtual currency which will ease the cash crisis since the economy would be using both the legal currency and the virtual currency; cryptocurrency technology is not exchanged in the black mark has a cure for the black

market problem that is prominent in the Zimbabwean economy; the technology can also be used as a way of settlement debt which eases the demand for cash and it offers financial inclusion to this excluded in the banking sector which also eases the demand of cash in the economy. Therefore an adoption of cryptocurrency has a negative effect on the liquidity crunch ($\beta = -0.115$, $p\text{-value} < 0.01$) hence the study concludes that:

H₂- There is a negative relationship between cryptocurrency adoption and liquidity crunch

The study analysed the effect of cryptocurrency adoption and the cost of remittances in the economy, Cryptocurrency adoption is believed to lower the cost of remittance in the economy. According to Team (2018) on an average, cryptocurrencies charge 0.04% of the total transaction as compared to a bank transfer which change around 10 to 15% depending on whether if it is a local or an international transfer. If the country adopts the cryptocurrency technology, it would result in cheaper remittances for both the migrants, business and online shoppers thus lowering the cost of transaction. This therefore means that if the economy adopts cryptocurrency, it result in the lowering of the cost of remittances thus a negative relationship emanates ($\beta = -0.334$, $p\text{-value} < 0.01$) thus the study concludes that:

H₃- There is a negative relationship between cryptocurrency adoption and cost of remittances

The study also found a positive relationship between the adoption of cryptocurrencies and the economic variables. The economic variable construct include variables such as inflation, easier of transacting, fast transactions, financial inclusion and divisibility of the currency. It is argued that due to the nature of the cryptocurrency technology which is a peer to peer type of network, there are no government pressures in the currency thus it is not affected by inflation. The technology offers easier of use and fast transaction to be made, it has the potential of including the financially excluded part of the society and they are no currency divisibility issues since it is highly divisible of up to 8 decimal places. Therefore the introduction of cryptocurrencies in the economy increases the economic variables ($\beta = 0.291$, $p\text{-value} < 0.01$) therefore the study conclude that:

H₄- There is a positive relationship between cryptocurrency adoption and economic variables

4.6 Summary

The chapter provided a detailed presentation, discussion and analysis of the research findings. The study investigated the effect of the adoption of cryptocurrency on the modern economy. The findings showed that government revenue and economic variables had a positive relationship with cryptocurrency adoption whilst liquidity crunch and cost of remittances have a negative relationship to cryptocurrency adoption. The next chapter looks at the summary of major findings, study conclusions and recommendations.

CHAPTER V

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.0 Introduction

The chapter focuses on the summary of the research, conclusion and recommendations based on the research findings. The chapter also provides the possible improvements basing on the shortfall observed and area of further study.

5.1 Summary

The research sought to achieve the following objectives

- **Objective 1:** *To explore the benefits of cryptocurrencies in the economy*
- **Objective 2:** *To investigate the regulatory framework that needs to be put in place for crypto currency operation in Zimbabwe*
- **Objective 3:** *To find out if cryptocurrency can solve the liquidity crunch and the higher cost of international remittances*
- **Objective 4:** *To analyse the opportunities and threats that cryptocurrencies poses to the economy*

5.2 Conclusions

The purpose of this study was to analyse the effect of cryptocurrency adoption on the modern economy. The study concluded using the objective as follows:

- **Objective 1:** *To explore the benefits of cryptocurrencies in the economy*

The findings that were obtained show that the adoption of cryptocurrency in the economy have potential benefit to the economy. There are many benefits and uses that cryptocurrency brings to the modern economy. These include the virtual currency being used a medium of exchange, it stops inflation, the system is not affected by network interruptions, easy of registration easier to use. The technology is highly divisible unlike the traditional currency which is divisible to a certain amount, cryptocurrency is divisible up to 8 decimal places. Cryptocurrency usage in an economy stops inflation because the maximum coins are strictly limited to certain number, there are neither political forces nor corporation that are able to change that order and it is

believed to financially include the excluded part of the society in financial service sector since it is easy to register and use. Cryptocurrency have an effect of increasing government revenue through taxation. If the government tax cryptocurrency on the purchase of the goods using the currency nor on ICOs would increase the government fiscals.

The study analysed the effect of the adoption of cryptocurrency and government revenue. From the analysis there was a positive relationship which means if the economy adopts cryptocurrency, the government can be able to gain revenue through taxes ($\beta = 0.121$, p-value < 0.01). Therefore from the results, the study concludes that cryptocurrency adoption has a positive effect to the modern economy.

- **Objective 2:** *To investigate the regulatory framework that needs to be put in place for crypto currency operation in Zimbabwe*

Most of the government globally have legalised cryptocurrency as a virtual currency rather than a legal tender. Governments argue that cryptocurrency is decentralised and does not need a central authority to regulate it, thus the government has its sore responsibility of issuing a legal currency and controlling the money supply. The respondents on average agreed ($\bar{x}=4.1033$ and $\sigma=0.23323$), that the government have to legalize cryptocurrency usage. The government have to allow cryptocurrency to operate as a virtual currency that can be allowed for the buying and selling of good. From the findings, the government have to adjust laws of the Control Financial Crime and Terrorism act so as to treat virtual currencies as a money service business and not be recognised as financial crime and terrorism. Some of the respondents agreed that the government have to make way for regulation that govern ICOs through reviewing the Securities Laws.

- **Objective 3:** *To find out if cryptocurrency can solve the liquidity crunch and the higher cost of international remittances*

The cryptocurrency adoption in the economy has been found to solve the liquidity crisis that is rampant our economy. The recognition of cryptocurrency as a virtual currency has allowed it to be used as a medium of exchange and as a method of settlement of debt thereby easing the cash demand. Therefore the providers of goods and services have to also start accepting cryptocurrency as a payment. Most governments like Canada and Switzerland are taking a lead by making the public pay for its services using cryptocurrency. This therefore reduces the cash crisis. The government have also has to encourage banks to include this technology in their

service offerings, Like Britain, they have to be allowed the investment in Bitcoin ATMs. Cryptocurrency like other fiat currencies that are traded with an exchange rate, this technology cannot be traded like other currencies at a current exchange rate therefore it may solve the black market exchange rate that keeps inflating the country's currency. The adoption of cryptocurrency in the economy may also have the ability to lessen the effects of inflation and financially include those excluded in the financial sector. Therefore the study concludes that ($\beta = -0.115$, $p\text{-value} < 0.01$) therefore:

H₂- There is a negative relationship between cryptocurrency adoption and liquidity crunch

The study analysed the effect of cryptocurrency adoption and the cost of remittances in the economy, Cryptocurrency adoption is believed to lower the cost of remittance in the economy. According to Team (2018) on an average, cryptocurrencies charge 0.04% of the total transaction as compared to a bank transfer which change around 10 to 15% depending on whether if it is a local or an international transfer. If the country adopts the cryptocurrency technology, it would result in cheaper remittances for both the migrants, business and online shoppers thus lowering the cost of transaction. This therefore means that if the economy adopts cryptocurrency, it result in the lowering of the cost of remittances thus a negative relationship emanates ($\beta = -0.334$, $p\text{-value} < 0.01$) thus the study concludes that:

H₃- There is a negative relationship between cryptocurrency adoption and cost of remittances

- ***Objective 4: To analyse the opportunities and threats that cryptocurrencies poses to the economy***

The study also found a positive relationship between the adoption of cryptocurrencies and the economic variables. The economic variable construct include variables such as inflation, easier of transacting, fast transactions, financial inclusion and divisibility of the currency. It is argued that due to the nature of the cryptocurrency technology which is a peer to peer type of network, there are no government pressures in the currency thus it is not affected by inflation. The technology offers easier of use and fast transaction to be made, it has the potential of including the financially excluded part of the society and they are no currency divisibility issues since it is highly divisible of up to 8 decimal places. Therefore the introduction of cryptocurrencies in the economy increases the economic variables (($\beta = 0.291$, $p\text{-value} < 0.01$) therefore the study conclude that:

H4- There is a positive relationship between cryptocurrency adoption and economic variables

5.3 Recommendations

The adoption of cryptocurrency have proved to have a positive effect to the economy based on the study findings. As the new normal the government of the day have to recognise it and adopt it as a currency like mobile money services. The Reserve Bank of Zimbabwe and Ministry of finance have to reconsider the ban imposed on cryptocurrency trading on the basing of the study findings, therefore the study recommends that:

Firstly the government have to issue laws that governs cryptocurrency operation in the country. The Reserve Bank have to come up with a policy document that clearly mentions how the cryptocurrency are to be recognised (digital currency or virtual currency), how it has to be used in the economy, the guidance of how the initial coin offerings have to be done and the laws that covers international remittances. The government have to include the use of cryptocurrency as taxable transaction. Therefore the study recommend that the government has to create a ground for cryptocurrency usage in the economy and make adjustments on the laws. For example the government has to include Initial Coin Offerings as a way of raising funds in the Securities laws, recognition of cryptocurrency as a virtual currency under the Control and Financial Crime Terrorism act to so as to exclude cryptocurrency. The RBZ has to put a watch unit in its technology supervision department so as to control the trading of cryptocurrency to avoid the misuse of it like the Silk Road scenario.

The adoption of cryptocurrency in the economy have proved to show benefits that are critical in the economic activities. Like money it can be used as a virtual currency in the settlement of debt, it can solve the black market crisis rampant in the economy, does not have any inflation since its it not affected by political nor corporation forces, can increase government revenue through inflation and can include the financially excluded part of the population in the financial services sector since it's easier to register and use.

The cost of remittances have been reduced by around 7% if the money is remitted by blockchain technology rather than the normal money service remittance. This has therefore reduced the cost of remittance significantly. The government has to allow cryptocurrency remittances since a significant population of Zimbabweans reside in other countries. The remittance have to constantly be monitored to avoid financial and terrorism crimes.

The government has to allow the trading of cryptocurrency in the economy since it bring more opportunities than threats in the economy. Cryptocurrency adoption consists of a series of benefits which range from solving the liquidity crunch in the economy, solving the inflation problems, including the financially excluded part of the society, highly divisible currency, and easy to use, low cost and faster transactions. Thus the study recommends for cryptocurrency adoption.

5.4 Area of further study

Cryptocurrency has continued to attract attention and consequently substantial investment from business, consumer and the media. The understanding of the driver of consumer adoption of the technology is recommended. Therefore the study recommends a study on the understanding of consumer adoption to cryptocurrency usage. The study may look at the purpose of consumer adoption of cryptocurrency (transaction or investment), the factors that influence consumer behaviours intention to use cryptocurrency and the factors that influence customer usage of cryptocurrency technology.

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APPENDIX

APPENDIX 1: COVER LETTER



Dear Sir/Madam

I am a Master's in Finance student with the Great Zimbabwe University. In order to complete my Master programme, I am required to carry out a research on: **THE ADOPTION OF THE CRYPTOCURRENCIES AND ITS EFFECT IN THE MODERN ECONOMY: A CASE OF THE ZIMBABWEAN ECONOMY**

I would be most grateful if could spare some time to attend to this questionnaire and interviews. This questionnaire and interviews forms the primary data of the study and the researcher hopes to come up with discussion points and recommendations on the subject matter.

I would like to thank you in advance for participating or responding to the questions and for your most valuable information and time. Your responses will be treated in confidentiality

Yours faithfully

Gerald Hlabiso

APENDIX 2-Questionnaires for RBZ and Ministry of Finance Employees

The researcher is Master of Finance student at Great Zimbabwe University carrying out a research on the topic that reads: **THE ADOPTION OF THE CRYPTOCURRENCIES AND ITS EFFECT IN THE MODERN ECONOMY: A CASE OF THE ZIMBABWEAN ECONOMY**. This research is conducted in partial fulfilment of the Masters of Finance Degree. You are kindly requested to complete each and every question frankly and honestly. The information you are providing will be treated in strictest of confidence and used for academic purposes only. Please do not indicate your name, signature or personal details anywhere on this paper. Your co-operation is greatly appreciated.

Instructions of completion of the questionnaire

- Please place a tick in the box where applicable
- Give additional information in the spaces provided where required
- Complete all sections of the questionnaire

Section A: General Information

May you please specify your gender?

Sex	Tick
Male	
Female	

May you please specify the highest level of education you have attained?

Level of Education	Tick
Diploma	
Degree	
Masters	
PhD	
Other	

For how long have you been using the Ministry of Finance or RBZ?

Years	Ticks
Less than a year	
1-5	

6-10	
Above 10 years	

Section B: Knowledge of Blockchain Technology

Do you know how Blockchain Technology works? Please tick appropriately

I have knowledge of how blockchain technology works	
Have had of blockchain technology in passing	
I don't have knowledge of how blockchain technology works	

Which Blockchain Technology have you used in your experience?

Yes	
No	

What was the main motive of using the technology? Please tick appropriately

For buying goods online	
For local money transfers	
For international remittance	
For Bitcoin mining	

At a rate of 1 to 5 how do you rate your experience?

	1	2	3	4	5
1 for not satisfied, 2 for satisfied ,3 for neutral and – 5 for strongly satisfied					

Section B: Benefits of Cryptocurrency to the economy

Please rate on the scale of 1 to 5 the extent you Agree or Disagree to the following statements. Please use the following scale and tick appropriately

1-Strongly Disagree 2-Disagree 3-Neutral 4- Agree 5- Strongly Agree

		1	2	3	4	5
Do you believe that cryptocurrency can be used as unit of measure like traditional money?						

Cryptocurrencies is highly divisible unlike traditional currencies of up to 8 decimal places. Do you believe that it can solve the divisibility problem that is prone to happen with the use of money?						
Cryptocurrency does not have a central authority which means central authority in the network has no power to dictate the rules for owners of Bitcoin. Do you believe than decentralization transaction is more efficient?						
Cryptocurrency does not have a central authority which means central authority in the network has no power to dictate the rules for owners of Bitcoin. Do you believe than decentralization transaction is more efficient?						
Bitcoin also offers a peer to peer cryptocurrency network where there is no master server, which is responsible for all operations, thus the exchange of information (money) is between 2-3 or more software's clients. Do you think anonymity transaction is good?						
The company has put in place processes that make it easier for the customer to order and buy their products						
Even if some part of the network goes offline, the payment will continue to operate stable. Do you think cryptocurrency is the solution for system disruptions?						
Cryptocurrencies usage in the economy stops inflation. The maximum of coins is strictly limited to a certain number, there are neither political forces nor corporations that are able to change that order. Do you believe that cryptocurrency can solve inflation problem in the economy.						
Cryptocurrencies are easier to register and to use comparing to the account the procedure of opening an accounts in banks. Do you think cryptocurrencies provide financial inclusion to the unbanked members of the society?						

Section C: Creating a Regulatory Framework for Cryptocurrency

Please rate on the scale of 1 to 5 the extent you Agree or Disagree to the following statements. Please use the following scale and tick appropriately

1-Strongly Disagree 2-Disagree 3-Nuetral 4- Agree 5- Strongly Agree

Do you think the government have to recognise cryptocurrency as a legal currency or a virtual or digital currency?

	Tick
Legal currency	
Virtual currency	

	1	2	3	4	5
Should the government amend its tax laws so as to include cryptocurrency in the taxation system for example included in the Income Tax?					
Can the government allow the selling and buying of goods using cryptocurrency although not allowing it to be a legal tender					
The RBZ has to consider cryptocurrency trading and review its Crime and Terrorism Act to treat virtual currency as a money service business for the purpose of anti-money laundering law					
The government has to review on securities laws so as to include Initial Coin offering and blockchain fund by publishing guidelines on its regulatory treatment in order to lure foreign direct investments					
In order for the public to gain acceptance of the cryptocurrency they have to make payment for their services with the option of Bitcoins					

Section D: Cryptocurrency, liquidity and cost of international remittances

Please rate on the scale of 1 to 5 the extent you Agree or Disagree to the following statements. Please use the following scale and tick appropriately

1-Strongly Disagree 2-Disagree 3-Neutral 4- Agree 5- Strongly Agree

Liquidity Concerns	1	2	3	4	5
If cryptocurrency is recognised as a virtual currency can be able to solve the liquidity crunch					
The government have to accept cryptocurrency as a way of payment and settlement of debts thus people would not demand more of the hard currency					
The government RBZ have to encourage the banks to include cryptocurrency services as well as ATMs.					
Cryptocurrency cannot be traded like other currencies at an exchange rate. Thus cryptocurrency can solve the money changing problem in the economy					
Most of the rural populace in Zimbabwe. Cryptocurrencies provide users easier registration and lower operating costs unlike physical cash there is no need to pay commission and fees to banks.					

The cost of remittance	1	2	3	4	5
The cost of remittance is high using financial services as compared to cryptocurrencies. Should the government allow cryptocurrencies remittances since there cost less					
Reducing the cost of remittances would results in more money remaining in the pockets of the migrant families and would have a significant effect on the income levels of remittance families					
Many migrants have faced difficulties to obtain the necessary identification documentation to enter the financial mainstream thus facing difficulties in sending remittances. Cryptocurrency has become the answer to this problem, it offers a new remittance solution that is significantly faster, cheaper and easier than the traditional model thus it must be adopted					
An increase in consumption, savings and investments in the local economies would lead to economic growth					
Most of the rural populace in Zimbabwe. Cryptocurrencies provide users easier registration and lower operating costs unlike physical cash there is no need to pay commission and fees to banks.					

Section E: Opportunities and threats of cryptocurrencies to the economy

Please rate on the scale of 1 to 5 the extent you Agree or Disagree to the following statements. Please use the following scale and tick appropriately

1-Strongly Disagree 2-Disagree 3-Nuetral 4- Agree 5- Strongly Agree

Opportunities	1	2	3	4	5
Bitcoin can help the unbanked part of the community to have peer to peer transactions which was not possible with traditional banking technology					
Bitcoins technology allows for individuals to exchange currency without needing a third trusted party, like a bank, to oversee the transaction. All that is needed to use Bitcoin is a mobile phone.					
The government can make revenue through taxation if cryptocurrencies are taxed for example Income Tax, Value added tax.					
The government can treat cryptocurrency has a commodity like Gold. Cryptocurrencies could potentially become a big player in the commodity market.					

Bitcoin has strength by design to make it a viable currency that has elevated it in status over the years, more notably the fixed limit of Bitcoin that will exist, Bitcoin will be mined with diminishing returns					
Threats					
The value fluctuations that plague cryptocurrencies puts doubt in users, as well as investors					
Cryptocurrencies have also seen fraud and theft, generally due to faulty system setups by exchange companies.					

Section F: The extent to cryptocurrency adoption and economic growth

Please rate on the scale of 1 to 5 the extent you Agree or Disagree to the following statements. Please use the following scale and tick appropriately

	1	2	3	4	5
The country have to adopt cryptocurrency use as it pose more opportunities to the country than threats					
The country must not adopt cryptocurrency use as it pose more threats to the country than opportunities					

APENDIX 3- Interview Guide

- How many years have you take operating in blockchain technology?
- What do most of your customers use cryptocurrency for?
- Do you also offer Altcoins to your customers
- What is your opinion on the benefit of cryptocurrency to the economy
- Do you think cryptocurrency can be registered by the government as a legal tender or digital currency
- Should the government amend its tax laws so as to include cryptocurrency in the taxation system for example included in the Income Tax?
- Can the government allow the selling and buying of goods using cryptocurrency although not allowing it to be a legal tender
- Do you think cryptocurrency can solve the liquidity crisis in the country
- Do you think that cryptocurrency can lower the cost of remittances
- In your opinion what are the opportunities that cryptocurrencies bring to the economy
- In your opinion what are the threats that cryptocurrencies bring to the economy
- Do you think cryptocurrency adoption increase economic growth